

LABORATORY MANUAL IN **DIGITAL ELECTRONICS 1:** **LOGIC CIRCUITS** **AND SWITCHING** **THEORY**

	Output
0	0
0	1
1	0
1	0



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PREFACE

This course covers design and analysis of digital circuits. This course covers both combinational logic circuits (synchronous and asynchronous), with an emphasis on solving digital problems using hard-wired structures for medium-to-large-scale integration.

This laboratory course is designed to provide hands-on experience introducing students to the applications and design of a priority encoder adder/subtractor, comparator, counter, multiplexer sequence generator and transform. Series 74H, LS, HS known to digital applications of integrated circuits. Students are made to design application circuits using basic integrated circuits. This would allow them to select the appropriate IC for the required application.

Course Objectives:

Analyze and design various applications using 7 segment display.

Design an 8 bit parallel load and serial out shift register.

Design digital circuits using 74H, LS, HS series IC's.

Design combinational logic circuits using digital ICs

Pre-requisite:

1. Students must passed the ECE 212 – Electronics 1 to enroll this subject.
2. Hence they will be having some of the knowledge regarding the Experiments in Electronics. In the current semester they will improve their skills by doing the experiments

Learning Outcomes:

By the end of the course, the students should be able to:

1. Cognitive
 - 1.1. Develop a digital logic and apply it to solve real life problems for industry , Innovation and Infrastructure(SDG-9) .
 - 1.2. Analyze, design, and implement combinational logic circuits.
 - 1.3. Classify different semiconductor memories.
 - 1.4. Analyze, design, and implement sequential logic circuits.
 - 1.5. Analyze digital system design using PLD in application to affordable and clean energy(SDG-7)
2. Psychomotor
 - 2.1. Display teamwork and commitment towards assigned tasks
3. Affective
 - 3.1 Use relevant software and hardware for the project Working of basic digital circuits

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Projects

Introduction

There are 3 hours allocated to a laboratory session in Digital Electronics. It is a necessary part of the course at which attendance is compulsory.

Here are safety guidelines to help you perform the experiments and to submit the reports:

1. Read all instructions carefully and carry them all out.
2. Ask a demonstrator if you are unsure of anything.
3. Record actual results (comment on them if they are unexpected!)
4. Write up full and suitable conclusions for each experiment.
5. If you have any doubt about the safety of any procedure, contact the demonstrator beforehand.
6. **THINK** about what you are doing!

Safety Instructions – Digital Laboratory

This document is to ensure the safe and smooth execution of the Digital Electronics 1 workshop. The safety rules have to be followed at all times when you are present in the teaching laboratories. Every student who participates in the lab course has to sign the safety agreement at the end of this document.

Emergency Response

1. It is your responsibility to read safety and fire alarm posters and follow the instructions during an emergency
2. Know the location of the fire extinguisher, eye wash, and safety shower in your lab and know how to use them.
3. Notify your instructor immediately after any injury, fire or explosion, or spill.
4. Know the building evacuation procedures

Common Sense

Good common sense is needed for safety in a laboratory. It is expected that each student will work in a responsible manner and exercise good judgement and common sense. If at any time you are not sure how to handle a particular situation ask your Instructor for advice. **DO NOT TOUCH ANYTHING WITH WHICH YOU ARE NOT**

COMPLETELY FAMILIAR!!! It is always better to ask questions than to risk harm to yourself or damage to the equipment

General lab guideline and safety instructions

1. Carry out the experiments in such a way that nobody will be injured or hurt.
2. Carry out the experiments in such a way that the equipment will not be damaged or destroyed.
3. Follow all written and verbal instructions carefully. If you do not understand the instructions, the hand-outs and the procedures, ask the instructor.
4. **Never work alone!** You should be accompanied by your laboratory partner and / or the instructors.
5. Perform only those experiments you find in the instructions or authorized by the instructors.
6. Unauthorized experiments are prohibited.
7. The workplace has to be tidy before, during and after the experiment.
8. Read the hand out and procedures before starting the experiments.
9. Intentional misconduct will lead to the exclusion from the lab.
10. Never eat, drink, or smoke while working in the laboratory.
11. Read labels carefully.
12. Do not use any equipment unless you are trained and approved as a user by your supervisor.
13. Wear safety glasses or face shields when working with hazardous materials and/or equipment.
14. Wear gloves when using any hazardous or toxic agent.
15. Clothing: When handling dangerous substances, wear gloves, laboratory coats, and safety shield or glasses. Shorts and sandals should not be worn in the lab at any time.
16. If you have long hair or loose clothes, make sure it is tied back or confined.
17. Keep the work area clear of all materials except those needed for your work. Coats should be hung in the hall or placed in a locker. Extra books, purses, etc. should be kept away from equipment, that requires air flow or ventilation to prevent overheating.
18. Disposal - Students are responsible for the proper disposal of used material.
19. Avoid contact with energized electrical circuits.
20. Use guarding around exposed circuits and sources of live electricity.
21. Disconnect the power source before servicing or repairing electrical equipment.
22. When it is necessary to handle equipment that is plugged in, be sure hands are dry and, when possible, wear nonconductive gloves and shoes with insulated soles.
23. If it is safe to do so, work with only one hand, keeping the other hand at your side or in your pocket, away from all conductive material. This precaution reduces the likelihood of accidents that result in current passing through the chest cavity.
24. Minimize the use of electrical equipment in cold rooms or other areas where condensation is likely. If equipment must be used in such areas, mount the equipment on a wall or vertical panel.
25. If water or a chemical is spilled onto equipment, shut off power at the main switch or circuit breaker and unplug the equipment.

26. If an individual comes in contact with a live electrical conductor, do not touch the equipment, cord or person. Disconnect the power source from the circuit breaker or pull out the plug using a leather belt.

Rules of conduct in case of emergency

First : keep cool

Second : warn persons at risk, punch the emergency shutdown

Third : inform the instructor

then : try to help, but take care of yourself

personal and General laboratory safety

1. Never eat, drink, or smoke while working in the laboratory.
2. Read labels carefully.
3. Do not use any equipment unless you are trained and approved as a user by your supervisor.
4. Wear safety glasses or face shields when working with hazardous materials and/or equipment.
5. Wear gloves when using any hazardous or toxic agent.
6. Clothing: When handling dangerous substances, wear gloves, laboratory coats, and safety shield or glasses. Shorts and sandals should not be worn in the lab at any time. Shoes are required when working in the machine shops.
7. If you have long hair or loose clothes, make sure it is tied back or confined.
8. Keep the work area clear of all materials except those needed for your work. Coats should be hung in the hall or placed in a locker. Extra books, purses, etc. should be kept away from equipment, that requires air flow or ventilation to prevent overheating.
9. Disposal - Students are responsible for the proper disposal of used material if any in appropriate containers.
10. Equipment Failure - If a piece of equipment fails while being used, report it immediately to your lab assistant or tutor. Never try to fix the problem yourself because you could harm yourself and others.
11. If leaving a lab unattended, turn off all ignition sources and lock the doors.
12. Never pipette anything by mouth.
13. Clean up your work area before leaving.
14. Wash hands before leaving the lab and before eating.

ELECTRONICS LABORATORY SAFETY

1. ALWAYS wear your SAFETY GLASS
2. KEEP soldering irons in their protective STAND when not in use.
3. Always CUT wire LEADS so the clipped wire falls on the table top and not toward others.
4. DO NOT TOUCH the tip end of a soldering iron to check for heat
5. AVOID on EARTH GROUND when working with AC powered units
6. ONLY work with powered units WHEN NECESSARY for troubleshooting

7. AVOID SKIN CONTACT with chemicals
8. REPLACE screws ,not just some.
9. Used the CORRECT CLEANING SOLVENTS for the JOB.
10. Avoid PINCHING wires when putting equipment back together.
11. Use a HEAT SINK when soldering temperature –sensitive components.
12. Never Solder a Circuit that's has the power applied.
13. Double check circuit for proper connections and polarity prior to applying power.
14. Observe POLARITY when connecting polarized components or test equipment into a circuit.
15. When soldering a multi-pin component .Avoid Excessive heating to one are of the component; DO NOT got for pin to pin in a straight line.
16. Make sure Test Instrument are set for proper FUNCTION and Range prior to taking measurement .
17. When measuring UNCERTAIN qualities ,start with the range switch on the HIGHEST setting .
18. Always REPLACE shield that were removed during service to avoid signal radiation .
19. When Cutting wires, Avoid cutting towards yourself.
20. Keep the INTENSITY on oscilloscopes as low as possible when in used and all the way down when not in use to avoid burning out the screen.
21. Always Observe Polarity when connecting components into a circuit ,especially with electrolytic capacitors.
22. Always READ the MSDS(Material Safety and Data Sheet).

Safety Instructions for the Digital Electronics 1

Agreement

I have read and agree to follow all of the safety rules. I realize that I must obey these rules to insure my own safety, and that of my fellow students, instructors and teaching assistants. I will cooperate to the fullest extent with my instructor and fellow students to maintain a safe lab environment. I will also closely follow the oral and written instructions provided by the instructor. I am aware that any violation of these rules that result in unsafe conduct in the laboratory or misbehavior on my part, may result in being removed from the laboratory, receiving a failing grade, and/or dismissal from the course.

Complete Name: _____

Course : _____ Date: _____

Signature : _____

Rubric for Assessment

Criteria	Unsatisfactory (1-3)	Satisfactory (4-6)	Good (7-9)	Excellent (10)
Completion (20%)	- less than 50% of the requirement has been completed	- between 50-65% of the requirement has been completed	- between 65-95% of the requirement has been completed	- between 100% of the requirement has been completed
Presentation and Organization (10%)	- no name, date, or requirement title included - poor use of materials - disorganized and messy	- name, date, requirement title included - use of materials that is easy to read - organized work	- name, date, requirement title included - good use of materials - organized work	- name, date, requirement title included - excellent use of materials - effective use of materials - creatively organized work
Correctness (30%)	- Circuit does not execute due to errors - no error checking code included - no testing has been completed	- Circuit executes without errors - Circuit handles some special cases - some testing has been completed	- Circuit executes without errors - Circuit handles most special cases - thorough testing has been completed	- Circuit executes without errors - Circuit handles all special cases - thorough and organized testing has been completed and output from test cases is included
Creativity (30%)	- Circuit uses a difficult and inefficient solution - designer has not considered alternate solutions	- Circuit uses a logical solution that is easy to follow but it is not the most efficient - designer has considered alternate solutions	- Circuit uses an efficient and easy to follow solution (i.e. no confusing tricks) - designer has considered alternate solution and has chosen the most efficient	- Circuit uses solution that is easy to understand and maintain - designer has analyzed many alternate solutions and has chosen the most efficient - designer has included the reasons for the solution chosen
Documentation (10%)	- no documentation included	- basic documentation has been completed	- program has been clearly documented	- clearly and effectively documented

The Breadboard

The breadboard consists of two terminal strips and two bus strips (often broken in the centre). Each bus strip has two rows of contacts. Each of the two rows of contacts are a node. That is, each contact along a row on a bus strip is connected together (inside the breadboard). Bus strips are used primarily for power supply connections, but are also used for any node requiring a large number of connections. Each terminal strip has 60 rows and 5 columns of contacts on each side of the centre gap. Each row of 5 contacts is a node.

You will build your circuits on the terminal strips by inserting the leads of circuit components into the contact receptacles and making connections with 22-26 gauge wire. There are wire cutter/strippers and a spool of wire in the lab. It is a good practice to wire +5V and 0V power supply connections to separate bus strips.

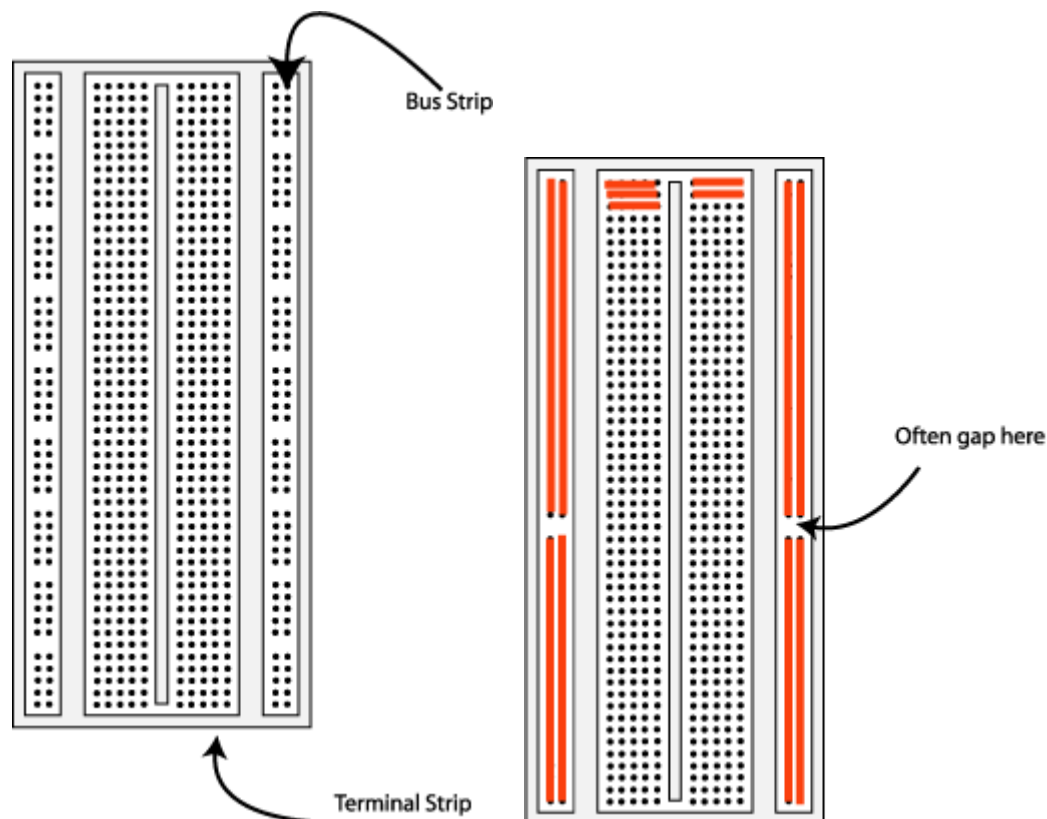


Fig 1. The breadboard. The lines indicate connected holes.

The 5V supply **MUST NOT BE EXCEEDED** since this will damage the ICs (Integrated circuits) used during the experiments. Incorrect connection

of power to the ICs could result in them exploding or becoming very hot - with the **possible serious injury occurring to the people working on the experiment! Ensure that the power supply polarity and all components and connections are correct before switching on power .**

Building the Circuit

Throughout these experiments we will use TTL chips to build circuits. The steps for wiring a circuit should be completed in the order described below:

1. Turn the power (Trainer Kit) off before you build anything!
2. Make sure the power is off before you build anything!
3. Connect the +5V and ground (GND) leads of the power supply to the power and ground bus strips on your breadboard.
4. Plug the chips you will be using into the breadboard. Point all the chips in the same direction with pin 1 at the upper-left corner. (Pin 1 is often identified by a dot or a notch next to it on the chip package)
5. Connect +5V and GND pins of each chip to the power and ground bus strips on the breadboard.
6. Select a connection on your schematic and place a piece of hook-up wire between corresponding pins of the chips on your breadboard. It is better to make the short connections before the longer ones. Mark each connection on your schematic as you go, so as not to try to make the same connection again at a later stage.
7. Get one of your group members to check the connections, **before you turn the power on.**
8. If an error is made and is not spotted before you turn the power on. Turn the power off immediately before you begin to rewire the circuit.
9. At the end of the laboratory session, collect you hook-up wires, chips and all equipment and return them to the demonstrator.
10. Tidy the area that you were working in and leave it in the same condition as it was before you started.

Common Causes of Problems

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1. Not connecting the ground and/or power pins for all chips.
2. Not turning on the power supply before checking the operation of the circuit.
3. Leaving out wires.
4. Plugging wires into the wrong holes.
5. Driving a single gate input with the outputs of two or more gates
6. Modifying the circuit with the power on.

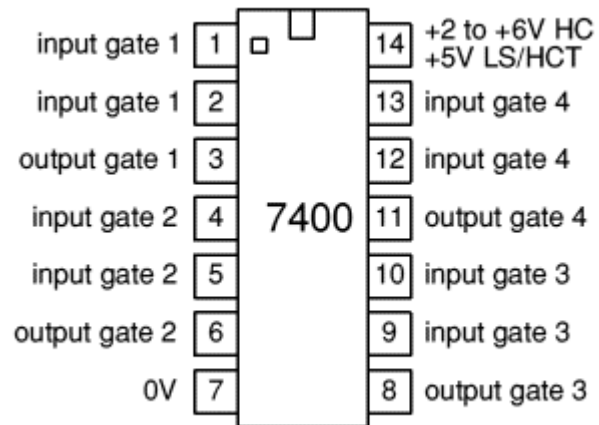
In all experiments, you will be expected to obtain all instruments, leads, components at the start of the experiment and return them to their proper place after you have finished the experiment. Please inform the demonstrator or technician if you locate faulty equipment. If you damage a chip, inform a demonstrator, don't put it back in the box of chips for somebody else to use.

If you locate any errors in this manual, please e-mail:-

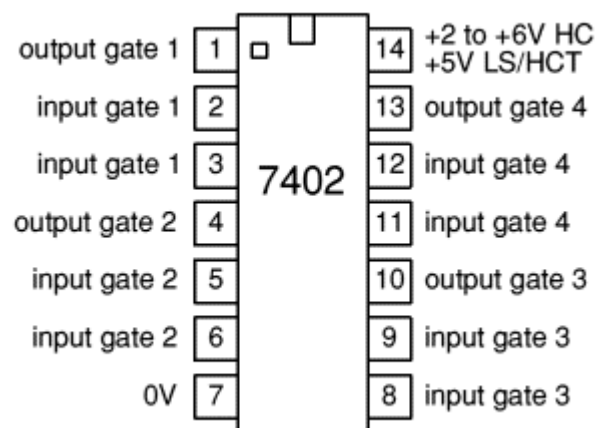
Jolan.b.sy@isu.edu.ph

Useful IC Pin details

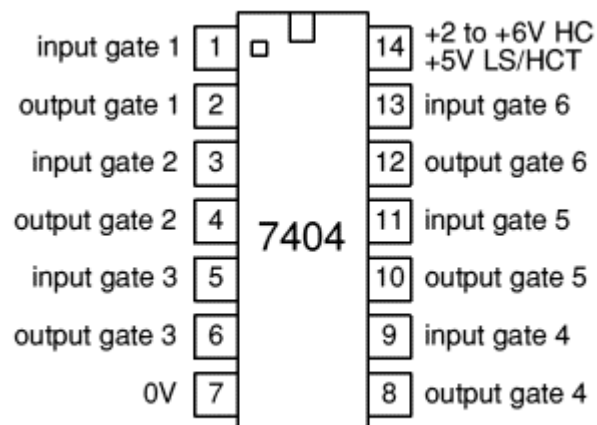
7400(NAND)



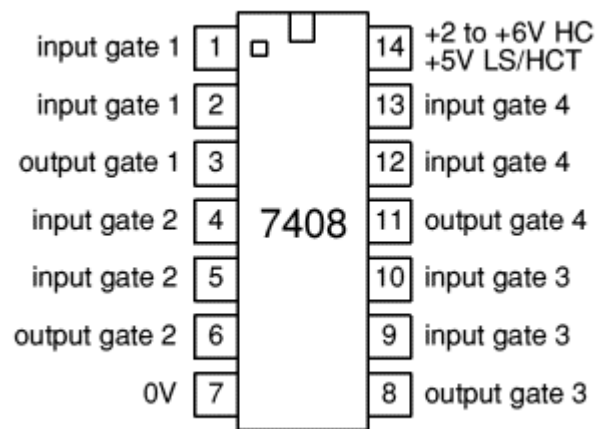
7402(NOR)



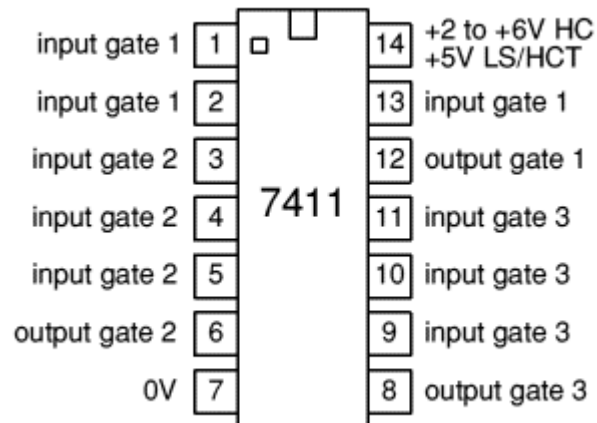
7404(NOT)



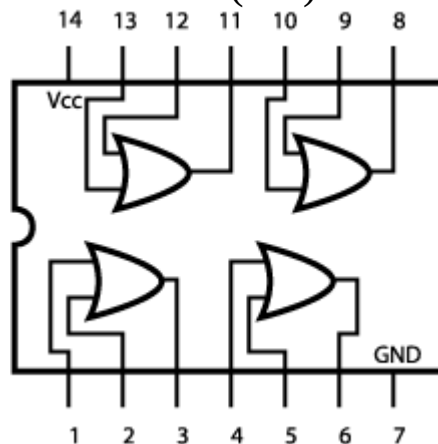
7408(AND)



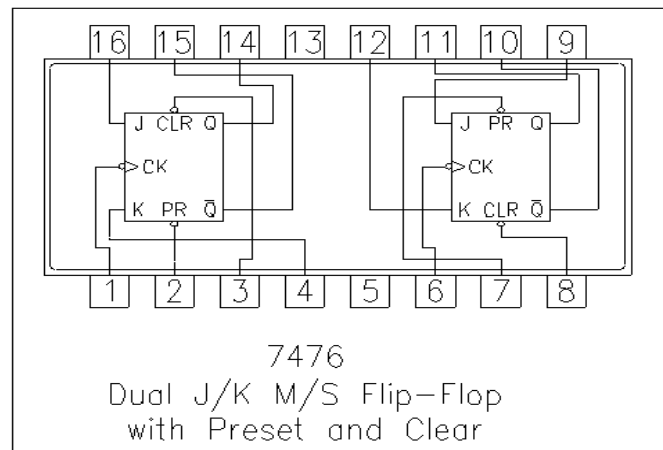
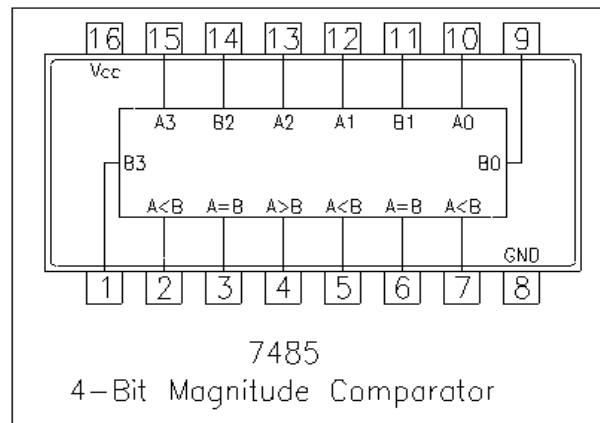
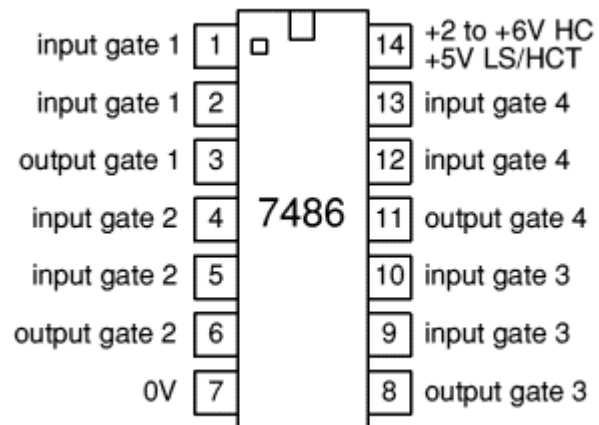
7411(3-i/p AND)



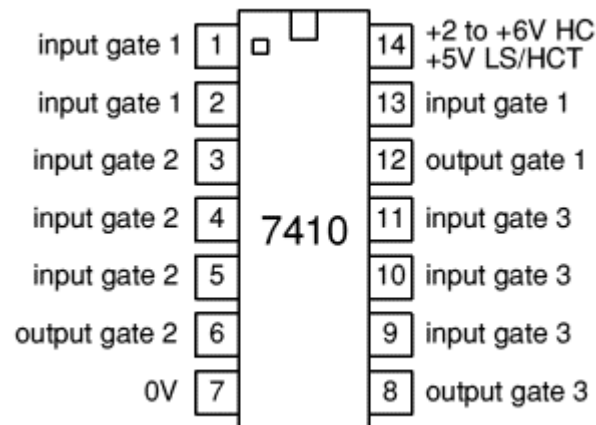
7432(OR)



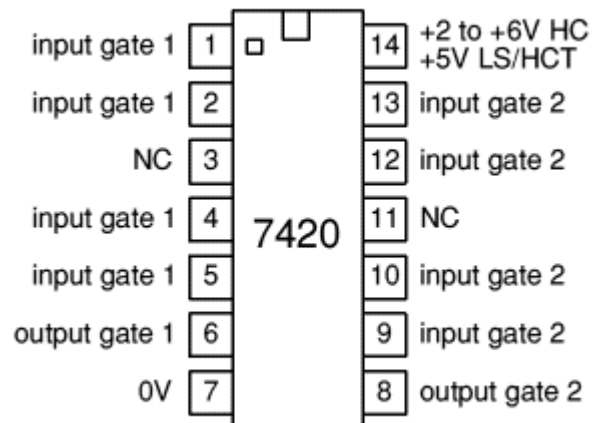
7486(EX-OR)



7410(3-i/p NAND)



7420(4-i/p NAND)



Experiment # 1

Rating: _____

Basic logic gates

Objective: At the end of the experiment, the students should be able to:

- Determine the basic function of Logic Gates & Switching theory.
- Illustrate the operations of basic Boolean Logic Gates.
- Implement basic binary logical operations using basic Boolean Logic Gates.

Materials:

IC# 7404 IC # 7408 IC # 7432 Connecting Wires Breadboard
5 volts DC Power Supply Logic Probe Light Emitting Diodes

Reminders:

Laboratory equipments & materials may cause harm to the researchers, please wear appropriate laboratory uniforms and protective gears for your safety. For ladies who have long hair and jewelries, it must be fixed to protect you from accidents

Theories:

The binary logic consists of binary variables and logical operations. The variables are designated by letters of the alphabet with each variable having two and only two distinct possible values: 1 and 0.

There are three basic logical operations: **AND**, **OR** and **NOT**.

1. **AND:** This operation is represented by a dot ($\bullet$) or by the absence of an operator. For example, $X \bullet Y = Z$ or $XY = Z$ is read as —X AND Y is equal to Z. The logical operation AND is interpreted to mean that $Z=1$ if and only if $X=1$ and $Y=1$; otherwise $Z=0$. (Remember that X, Y and Z are binary variables and can be equal to either 1 or 0, and nothing else.)

2. OR: This operation is represented by a plus (+) sign. For example, $X+Y=Z$ is read as

—X OR Y is equal to Z meaning that $Z=1$ if $X=1$ or $Y=1$ or if both X and Y is equal to 1. If both X and Y is equal to 0, then $Z=0$.

3. NOT: This operation is represented by a prime ($_$) or sometimes by a bar ($\bar{}$). For example, $X' = Z$ (or $X=Z$) is read, —X is not equal to Z. In other words, If $X=1$ then $Z=0$; but if $X=0$, then $Z=1$.

For each combination of the values of X and Y, there is a value of Z specified by the definition of the logical operation. These definitions may be listed in a compact form using **truth tables**. A truth table is a table of all possible combinations of the variables showing the relation between the values that the variables may take and the result of the operation.

Procedures:

1.] Using IC 7408, IC 7432, and IC 7404, respectively, and Light Emitting Diodes (LED), Connect the circuit shown in fig. a, fig B. & fig. C.

** Make sure that the Power supply is set to **+5 VOLTS only**.*

2.] Input the values stated in table A for figure A, table B for Figure B, and table C for figure C.

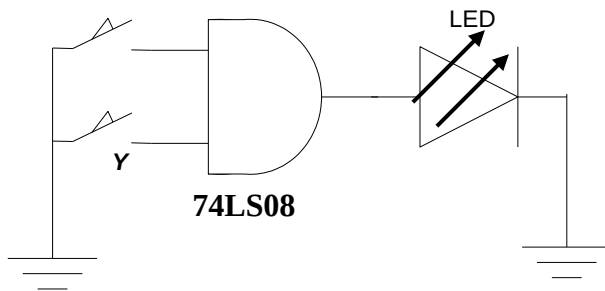
3.] Record the events made by the LED in the table.

Legend:

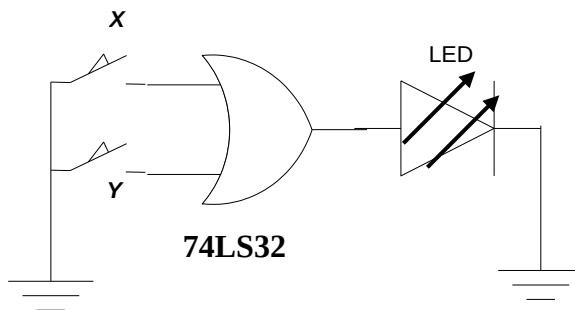
Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1
Output	Value Representation
Led —OFF	0
Led ON	1

Figure A:
x

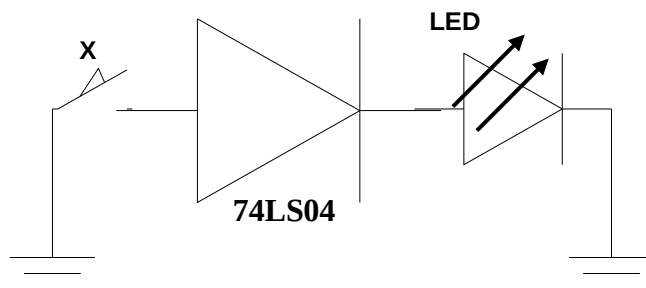
Table A:
Input Output

**Figure B:**

X	Y		LED
0	0		
0	1		
1	0		
1	1		

Table B:**Input****Output****Figure C:**

X	Y		LED
0	0		
0	1		
1	0		
1	1		

Table C:**Input****Output**

X		LED
0		
1		

4.] After implementing the Circuit above and recording the output in the table provided, write the equivalent Boolean equation for the three figures that you have implemented.

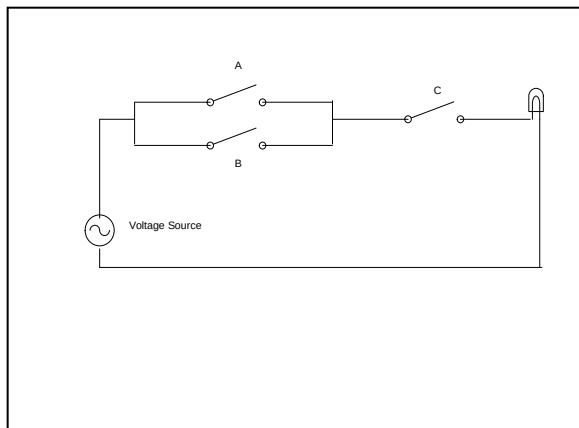
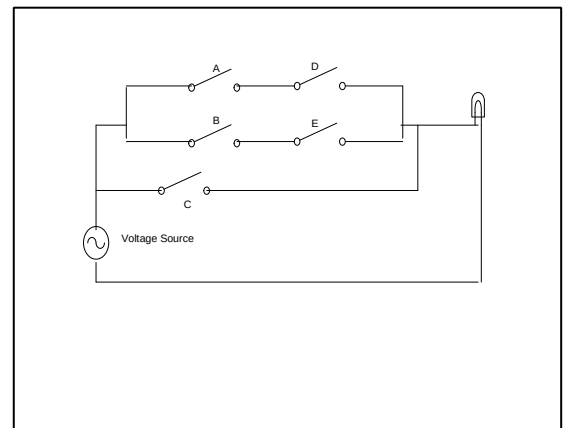
5.] Perform the activity using Software Simulation (Multisim/Proteus or Circuit Maker)

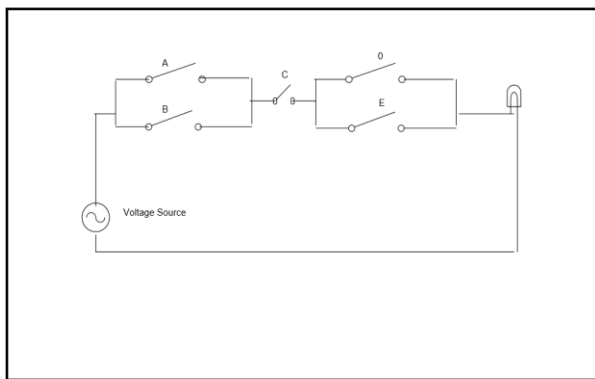
Conclusions/ Recommendations:

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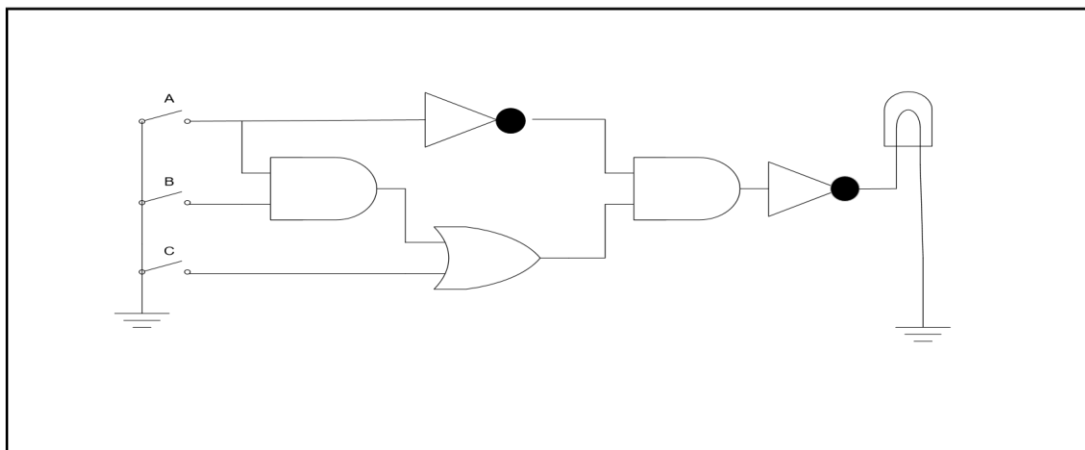
Problem:

1. What were the similarities noticed from the experiment that we can, or have applied from the Philosophy subject?
2. Express the following electrical circuits in Binary logic notation.

A.**B.**

C.

3. Draw the electrical circuit representation (same as above) the logic circuit figure shown below:



Experiment # 2**Rating:** _____**Universal Logic Gates**

Objective: At the end of the experiment, the students should be able to:

- Determine the basic function of Universal Logic Gates.
- Illustrate the operations of basic Boolean Universal Logic Gates.
- Implement basic binary logical operations using basic Boolean Universal Logic Gates.

Materials:

IC# 7400

IC # 7402

Connecting Wires Breadboard

5 volts DC Power Supply

Logic Probe

Light Emitting Diodes

Reminders:

Laboratory equipment's & materials may cause harm to the researchers, please wear appropriate laboratory uniforms and protective gears for your safety. For ladies who have long hair and jewelries, it must be fixed to protect you from accidents **Theories:**

Another type of logic gates that we can use are the **NAND** and **NOR** gates, also known as universal logic gates. This type of gates can also be used as alternates for AND, OR and NOT gates.

The NAND function is the complement of the AND function (AND with an Inverter connected in the output) and it's name is the abbreviation of *not-AND*, with symbol as shown by the graphic symbol which consist of a AND gate followed by a circle.

The NOR function is the complement of the OR function (OR with an Inverter connected in the output) and it's name is the abbreviation of *not-OR*, with symbol as shown by the graphic symbol which consist of a OR gate followed by a circle.

The NAND and NOR are extensively used as standard logic gates and are in fact far more popular than the AND and OR gates. This is because NAND and NOR gates are

easily constructed with transistor circuits and because Boolean functions can be easily implemented with them.

Procedures:

1.] Using IC 7400 and IC 7402, respectively, and Light Emitting Diodes (LED), implement the circuit shown in Figure A. and Figure B.

**make sure that the Power supply is set to +5 VOLTS ONLY*

2.] Input the values stated in table A.

Legend:

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the events made by the LED in the table.

Legend:

Output	Value Representation
Led —OFF	0
Led ON	1

Table A

INPUT		Output	
X	Y		LED
0	0		
0	1		
1	0		
1	1		

Figure A:

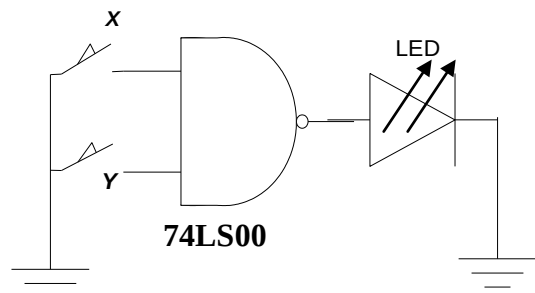
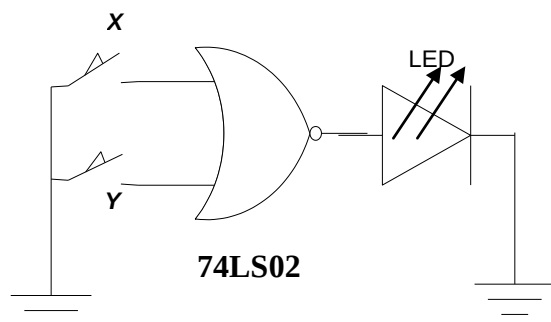


Table B

INPUT		Output	
X	Y		LED
0	0		
0	1		
1	0		
1	1		

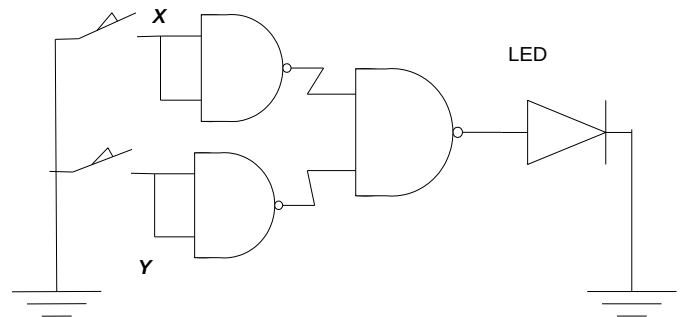
Figure B:



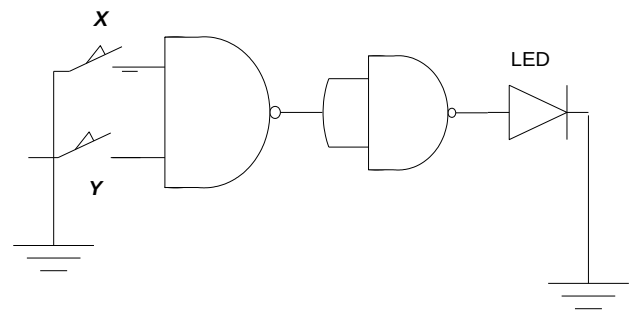
4.] Using a similar procedure, implement the circuit shown in Figures C, D and E. Then record the events in their corresponding tables.

Table C

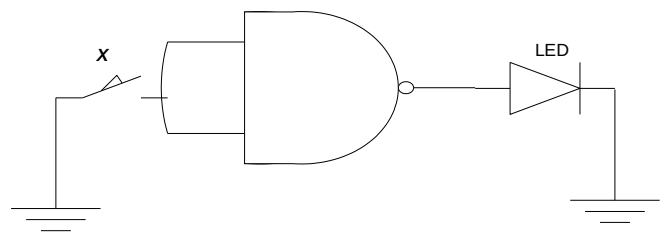
INPUT		Output	
X	Y		LED
0	0		
0	1		
1	0		
1	1		

Figure C:**Table D**

INPUT		Output	
X	Y		LED
0	0		
0	1		
1	0		
1	1		

Figure D:**Table E**

INPUT		Output
X		LED
0		
1		

Figure E:

5.] Implement another circuit shown in Figure F, G and H using the same procedure as above. Then record the events in their corresponding tables.

6.] Perform the circuit using any simulation software and compare the results.

Table F

INPUT		Output	
X	Y		LED
0	0		
0	1		
1	0		
1	1		

Figure F:

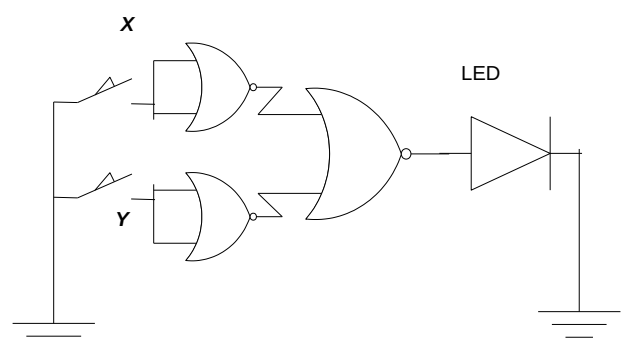


Table G

INPUT		Output	
X	Y		LED
0	0		
0	1		
1	0		
1	1		

Figure G:

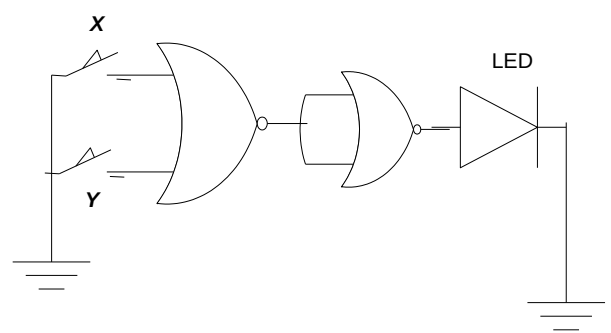
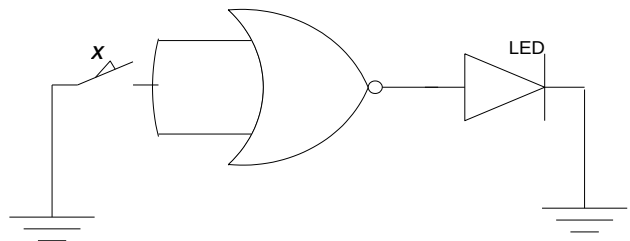


Table H

INPUT		Output
X		LED
0		
1		

Figure H:

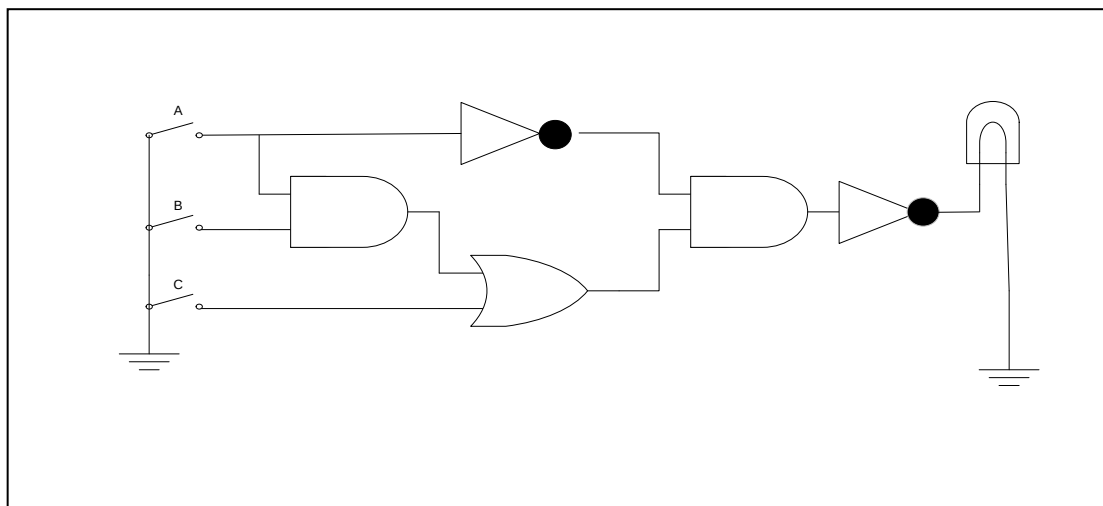


Conclusions/ Recommendations:

[illegible]

Test Problem:

1. What would be the advantages and disadvantages of using Universal gate as a replacement for the Basic gates used in Experiment #1?
2. Show the Electrical circuit (by means of a diagram similar to the previous experiment) of the Logic circuits in Figure C, D, F, and G.
3. If we were to replace the Basic gates shown in figure 2.a with any universal gate, how many gates should we use?

**Fig. 2.a**

4. Assuming each gate would have a processing time (From entering Input signals to receiving output signals) of 50 nSec:
 - a. How much time would be needed to produce the output (Light the LED) in Fig.2.a?
 - b. How much additional time would it produce using universal gates?
5. In substituting the Universal gate for the Basic gates, can we eliminate unnecessary gates used during substitution? How and Why?

Experiment # 3**Rating:** _____**Boolean Expressions****Objective:** At the end of the experiment the students should be able to:

- Apply the Boolean principles in logic diagram or circuits through Boolean expression implementation.
- Appreciate Boolean principles and expression in logic diagram or circuits

Materials:

IC# 7404 IC # 7408 IC # 7432 Connecting wires IC # 7486
5 volts DC Power Supply Logic Probe Light Emitting Diodes

Breadboards Reminders:

Laboratory equipment's & materials may cause harm to the researchers, please wear appropriate laboratory uniforms and protective gears for your safety. For ladies who have long hair and jewelries, it must be fixed to protect you from accidents

Theories:

The best way to familiarize the Boolean principles is by obtaining the equivalent logic circuit from a given Boolean expression. The process requires mastery of the different gate characteristics and analysis on the possible output for a given set of hypothetical input signals. Also, the circuit designer or analyst should apply different steps in obtaining the circuit. These are as follows:

1. Get the logic diagram (circuit) of the Boolean expression.
2. Obtain the truth table for the circuit.

Note: Make sure that all possible combinations and their corresponding outputs are correct. And for him to be assured that the circuit being derived is correct, he has to physically connect or construct the circuit.

Theories:

1.] Using IC 74series and Light Emitting Diodes (LED), implement the logic diagram of the given Boolean expressions below:

- *Make sure that the Power supply is set to **+5 VOLTS ONLY**.
- * If an XOR gate is not available in the laboratory; you will have to expand the equation using alternate gates or circuit (i.e. Universal gates or Basic gates).

Boolean expressions:

A:

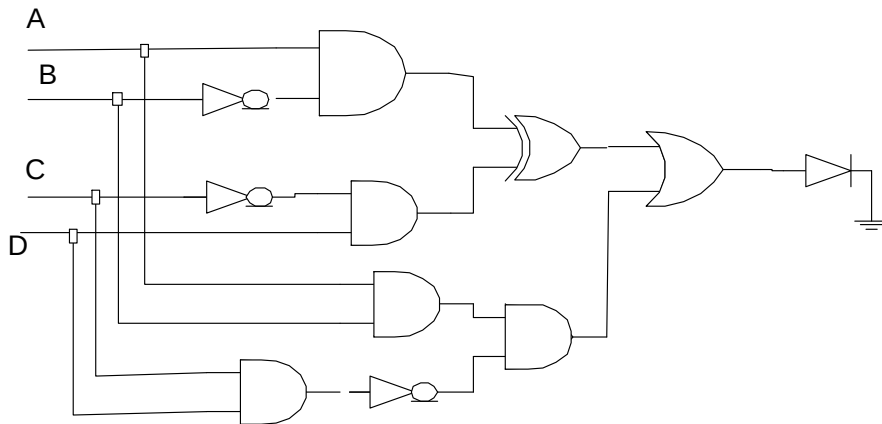
$$F(A, B, C, D) = (\bar{A}B \oplus \bar{C}D) + AB\bar{C}\bar{D}$$

B:

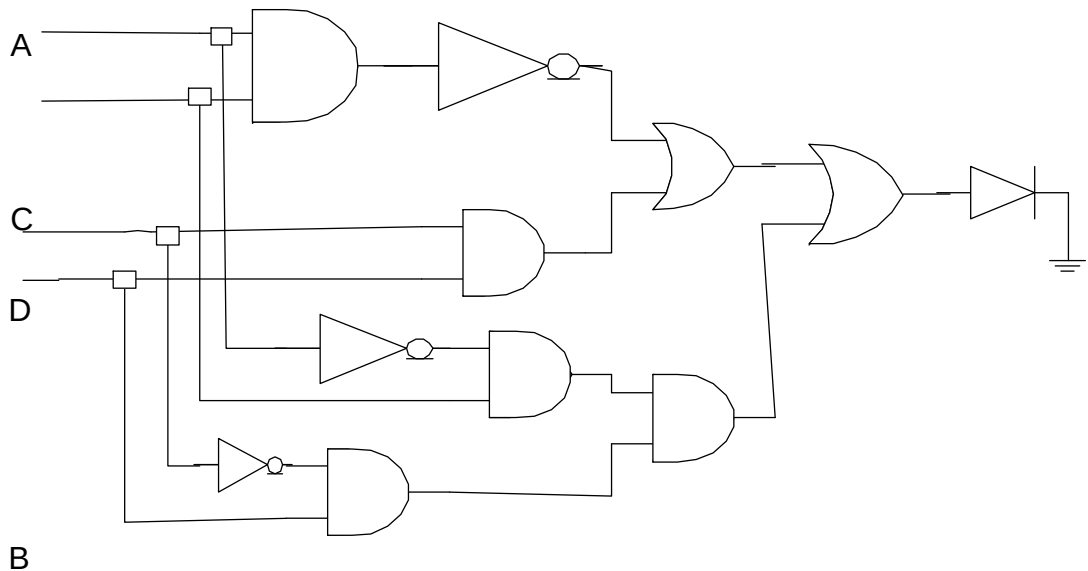
$$F(A, B, C, D) = (\bar{A}\bar{B} + CD) + A\bar{B}\bar{C}\bar{D}$$

Logic Diagram:

A:



B:



2.] Trace the circuit and make a truth table then put the correct input combinations and their respective theoretical outputs. Allocate another column for the practical/physical outputs, which will be filled later upon Test the circuit with the same inputs.

3.] Implement the Boolean expressions using a different equipment and device available.

4.] Perform the Circuit using any simulation software and compare the results.

Conclusions/ Recommendations:

Test Problem:

1. In the experiment, should a designer consider the number of standard loads, also known as Fan-out, the output of a gate can drive without impairing its normal operations?
2. In an algebraic expression converted into Logic circuits, should we bother on propagation delays?
3. If gates were replaced in a Universal gate counterpart, the logic diagram will be modified. Will it affect the processing speed and cost of the overall circuit?
4. When implementing Logic circuit diagrams, is it advisable to use a NAND or NOR gates than the OR, AND, or NOT gates?
5. Would it be easier to implement Boolean equations into Logic circuits starting from the left variable to the right? Or from variables enclosed in parenthesis, brackets, or with Prime or bar?

Experiment # 4**Rating:** _____**Boolean expression simplification****Objective:** At the end of the experiment the students should be able to:

- Apply the Karnaugh - map principle in obtaining a Boolean expression from a given truth table.

- Simplify equations using Karnaugh - map principle

Materials:

IC# 7404	IC # 7408	IC # 7432	Connecting Wires
IC # 7402	IC # 7486	IC # 7400	Light Emitting Diode
5 Volts DC Power Supply		Logic Probe	Breadboard

Reminders:

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Theories:

In Boolean algebra, there are different properties/laws/theorems to simplify either a given Boolean expression or one obtained from a truth table. Also, there are different techniques designed for Boolean simplification.

In most respects, these different principles of Boolean simplification are the ones responsible of why today there are a lot of different computer design and structures. One of the most popular techniques in this area of computer system is the Boolean simplification by map method.

More widely known as the Karnaugh - mapping, the principle was introduced by Veitch and perfected by karnaugh. This is also called as Veitch diagram.

This Process involves drawing a map that is based on the relationship between the input and output signals of a circuit. The size/dimension (number of cells) of the map is obtained by the formula 2^n , where n is the number of inputs.

Example:

Number of inputs (n) = 2. Therefore map size = $2^2 = 4$ cells.

Y \ X	0	1
0		
1		

* For 3 input, map size = 8 cells

* For 4 input, map size = 16 cells

* For 5 input, map size = 32 cells

Each cell in the map represents a single minterm (SOP). The main reason why Kmap method is the most commonly used in Boolean simplification is that whatever Boolean format is obtained from the process, there is almost no further simplification needed.

Procedures:

1.] Obtain the simplified Boolean expression of the logic circuit represented by the truth table below. Use the K-map method.

Truth table:

INPUT				OUTPUT			
A	B	C	D	O4	O3	O2	O1
0	0	0	0	1	0	0	1
0	0	0	1	0	0	1	1
0	0	1	0	1	0	1	1
0	0	1	1	0	0	1	1
0	1	0	0	1	1	1	0
0	1	0	1	0	1	1	0
0	1	1	0	1	1	1	0
0	1	1	1	1	1	1	0
1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	0
1	0	1	0	1	0	0	0
1	0	1	1	1	0	0	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	0
1	1	1	0	1	1	1	0
1	1	1	1	1	0	1	0

2.] Show the simplification process/solution below. (* Each output (**O4, O3, O2 & O1**) has an individual Boolean expression)

3.] Draw the simplified logic circuit of the simplified equation obtained using K-map.

4.] Obtain the un-simplified equation and implement it's circuit using the standard implementation (un-simplified). Compare the circuits, Simplified and not, with the results stated in the table above.

***Make sure that the Power supply is set to +5 VOLTS ONLY.**

Conclusions/ Recommendations:

[illegible]

Test Problem:

1. Why is the arrangement of the Karnaugh map diagram's boxes not in sequential order?
2. Can Karnaugh map be modified to further enhance simplification of Boolean equations?
3. Is the simplification using Karnaugh map be as flexible and as effective as the standard Theorems and Postulates?
4. Since Karnaugh map is used in simplifying logic in circuits, would it be possible to apply Karnaugh map in the real world situations?
5. Would it be as effective as the traditional decision-making techniques?

Experiment # 5

Rating: _____

Combinational Logic Circuit: Complements

Part 1: (R-1)'s Complement Circuit

Objective: At the end of the experiment, the students should be able to:

- Implement physically the principle of complement used in Binary Subtraction.
- Evaluate the result of the operation with respect to efficiency & Correctness.

Materials:

IC# 7404	Connecting Wires	5 Volts DC Power Supply	Logic Probe
Breadboard	Light Emitting Diodes		

Reminders:

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Theories:

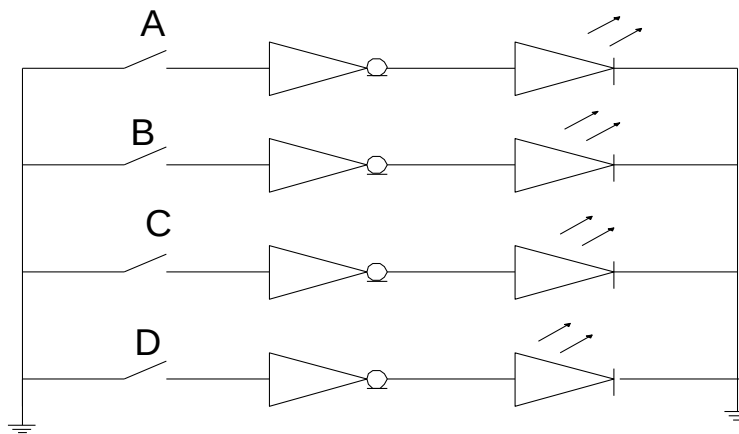
One way of obtaining the difference between 2 binary numbers (binary subtraction) is through the application of the 1's complement technique or principle. Theoretically, the 1's complement of a binary number (subtrahend) may be obtained by simply getting the opposite states (1 or 0) of each of its bits.

Physically, 1's complement circuit may be constructed by connecting a not gate after the input components (may be a switch) that represents a binary bit.

Procedures:

1.] Using IC 7404 and Light Emitting Diodes (LED), connect the circuit diagram below:

*Make sure that the Power supply is set to +5 VOLTS ONLY



2.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

Output	Output Value
Led —OFF	0
Led ON	1

Truth table:

INPUT				THEORETICAL				IMPLEMENTED			
A	B	C	D	O4	O3	O2	O1	B4	B3	B2	B1
0	0	0	0								
0	0	0	1								
0	0	1	0								
0	0	1	1								
0	1	0	0								
0	1	0	1								
0	1	1	0								
0	1	1	1								
1	0	0	0								
1	0	0	1								
1	0	1	0								
1	0	1	1								
1	1	0	0								
1	1	0	1								
1	1	1	0								
1	1	1	1								

PART 2: (R)'s Complement Circuit

Theories:

One way of obtaining the difference between 2 binary numbers (binary subtraction) is through the application of the 2's complement technique or principle. Theoretically, the 2's complement of a binary number (subtrahend) may be obtained by simply getting the opposite states (1 or 0) of each of its bits and add a binary 1 to its 1's complement.

Physically, 2's complement circuit is not as simple as the 1's complement, since this requires not only analysis from the user but also the application of different logic gate principles.

Procedures:

1.] Connect the circuit given (separate sheet).

*Make sure that the Power supply is set to **+5 VOLTS ONLY**

** Make sure that the IC's pin configurations are correctly followed in connecting the different circuit components.

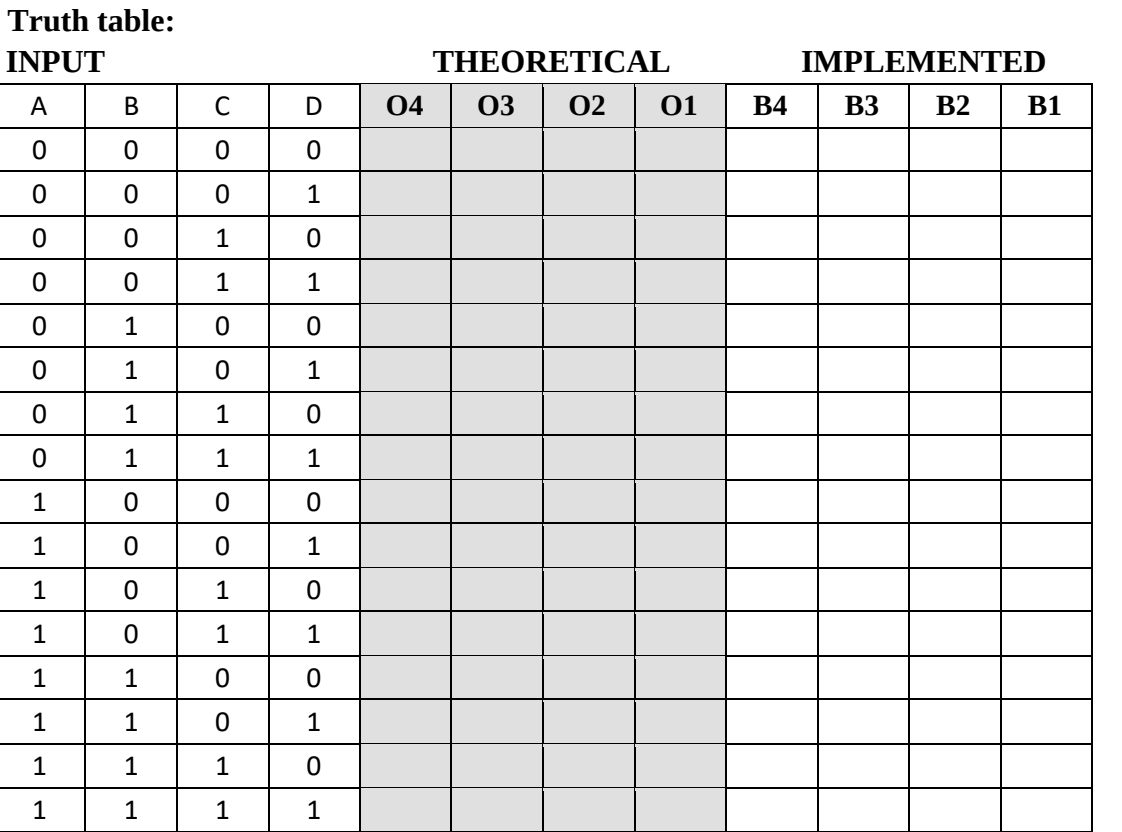
2.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

Output	Output Value
Led —OFF	0
Led ON	1

Logic diagram:

[illegible]

Experiment # 6**Rating:** _____

The Half Adder circuit

Objective:

At the end of the experiment, the students should be able to apply the combinational logic circuits (CLC) principles and process by implementing binary arithmetic circuits.

Materials:

IC# 7404	IC # 7408	IC # 7486	IC# 7432
Connecting Wires	5 Volts DC Power Supply		Logic
Probe Breadboard	Light Emitting Diodes	Reminders:	

Laboratory equipments & materials may cause harm to the researchers, please wear appropriate laboratory uniforms and protective gears for your safety. For ladies who have long hair and jewelries, it must be fixed to protect you from accidents

Theories:

One of the major units of the computer CPU is the arithmetic logic unit. In the software aspect, this area of the system handles all the logical and arithmetic operations of the computer. In the hardware standpoint, this area is composed of a network of circuits; mainly of gate IC's which are designed to manage arithmetic and logical processes. And one of these components is an Adder operator. Through this activity, a circuit analyst will be able, not only to understand these basic internal operations of the computer, but also to construct separate circuits for them.

A half -adder is a combinational logic circuit capable of adding 2 binary bits (Addend & Augend) whose outputs are the Sum and the Carry.

Procedures:

1.] Implement the circuit showed in fig. A, fig. B & fig. C. Input the values in the table. *Make sure that the Power supply is set to **+5 VOLTS ONLY**

2.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

Output	Value Representation
Led —OFF	0
Led ON	1

4.] From the given tables, obtain the equations of C and S.

Truth Table A:

X	Y	C	S
0	0		
0	1		
1	0		
1	1		

Input

Output

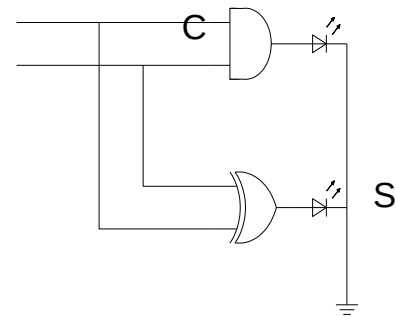


Figure A:

Truth Table B:

X	Y	C	S
0	0		
0	1		
1	0		
1	1		

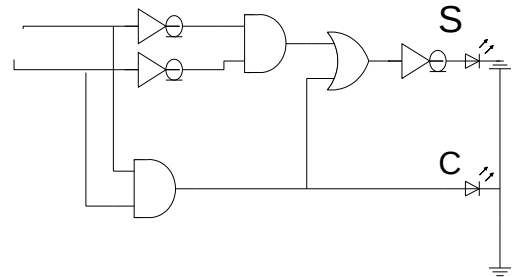


Figure B:

Truth Table C:

X	Y	C	S
0	0		
0	1		
1	0		
1	1		

Input

Output

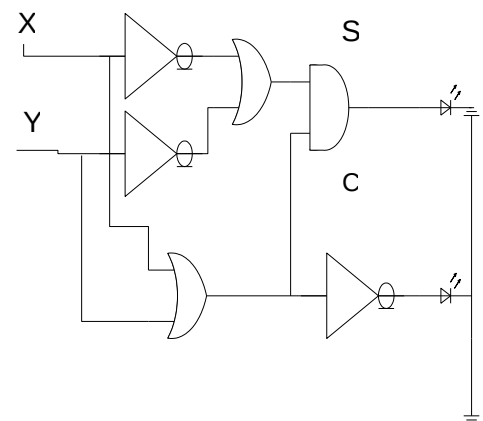


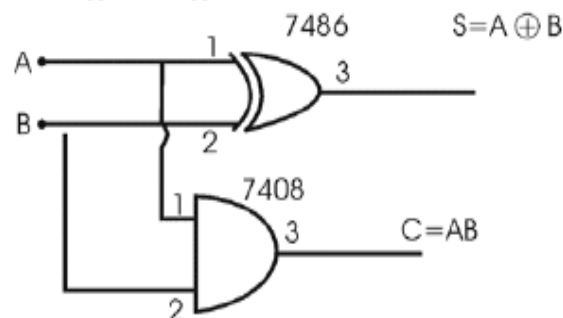
Figure C:

Conclusions/ Recommendations:

[illegible]

Test Problem

Half Adder using basic gates:-

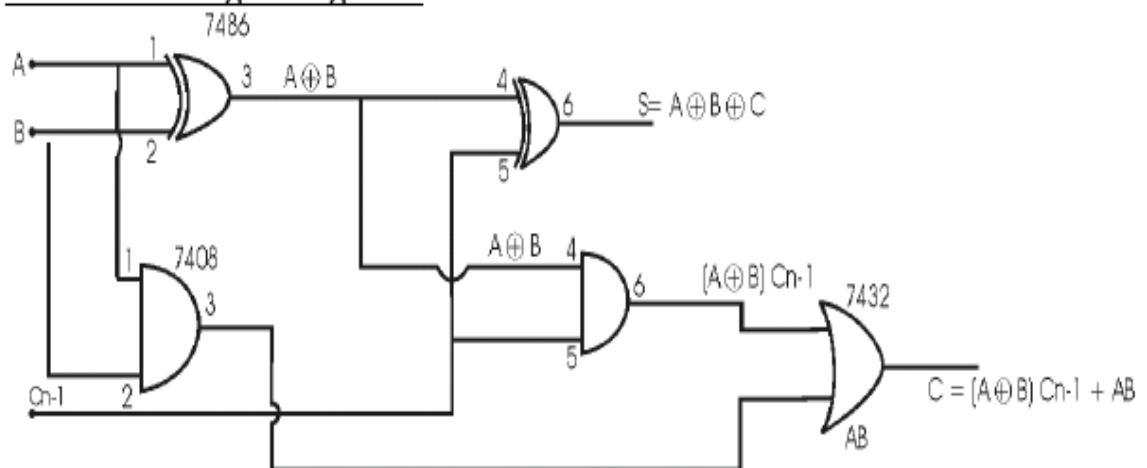


$$S = \overline{A}B + A\overline{B}$$

$$S = A \oplus B$$

$$C = AB$$

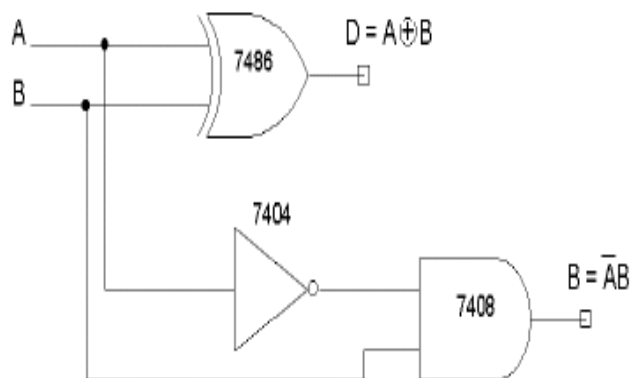
Full Adder using basic gates:-



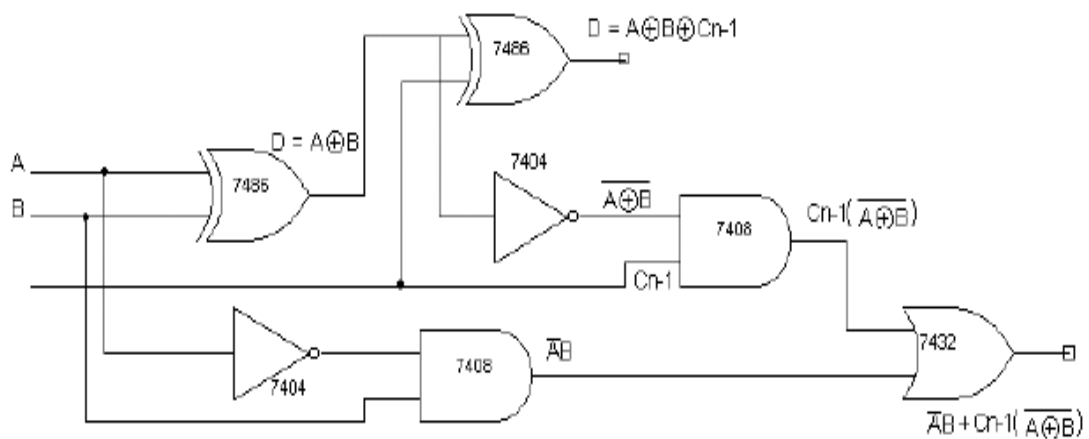
Half Adder					
A	B	S	C	S(V)	C(V)
	0	0	0		
0	1	1	0		
1	0	1	0		
1	1	0	1		

Full Adder						
A	B	C _{n-1}	S	C	S(V)	C(V)
0	0	0	0	0		
0	0	1	1	0		
0	1	0	1	0		
0	1	1	0	1		
1	0	0	1	0		
1	0	1	0	1		
1	1	0	0	1		
1	1	1	1	1		

Using X – OR and Basic Gates (a)Half Subtractor



Full Subtractor



Half Subtractor					
A	B	D	B	D(V)	B(V)
0	0	0	0		
0	1	1	1		
1	0	1	0		
1	1	0	0		

Full Subtractor						
A	B	C _{n-1}	D	B	D(v)	B(v)
0	0	0	0	0		
0	0	1	1	1		
0	1	0	1	1		
0	1	1	0	1		
1	0	0	1	0		
1	0	1	0	0		
1	1	0	0	0		
1	1	1	1	1		

Experiment # 6B**Rating:** _____**The Half Subtractor Circuit****Objective:**

At the end of the experiment, the students should be able to apply the combinational logic circuit principles and processes by implementing binary arithmetic circuits, particularly the half-subtractor circuit.

Materials:

IC# 7404 IC # 7408 IC # 7432 Connecting Wires Light Emitting Diodes
5 Volts DC Power Supply Logic Probe Breadboard

Reminders:

Laboratory equipment's & materials may cause harm to the researchers, please wear appropriate laboratory uniforms and protective gears for your safety. For ladies who have long hair and jewelries, it must be fixed to protect you from accidents

Theories:

One of the major units of the computer CPU is the arithmetic logic unit. In the software aspect, this area of the system handles all the logical and arithmetic operations of the computer. In the hardware standpoint, this area is composed of a network of circuits; mainly of gate IC's which are designed to manage arithmetic and logical processes. And one of these components is a Subtractor operator. Through this activity, a circuit analyst will be able, not only to understand these basic internal operations of the computer, but also to construct separate circuits for them.

A half - subtractor is a combinational logic circuit capable of subtracting 2 binary bits (Minuend & Subtrahend) whose outputs are the Difference and the Borrow.

Procedures:

1.] Using any available 74 IC series, implement the circuit shown in fig. A, fig. B & fig.

C

*Make sure that the Power supply is set to **+5 VOLTS ONLY**

2.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

Output	Value Representation
Led —OFF	0
Led ON	1

4.] From the given tables, obtain the equations of B and D.

Truth Table A:

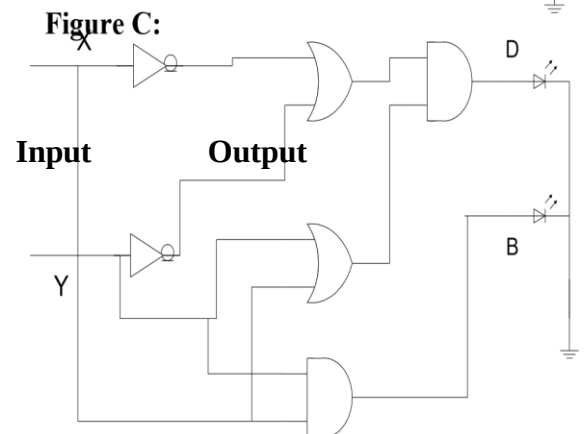
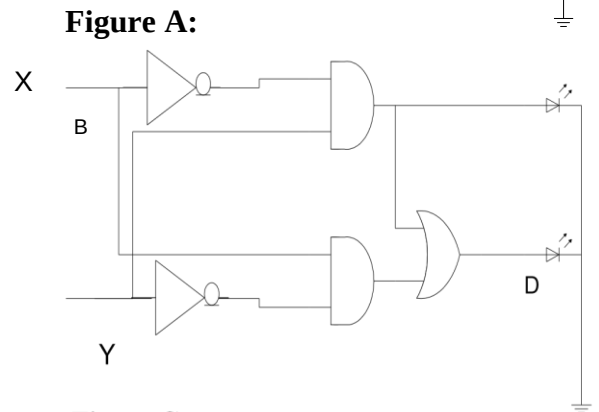
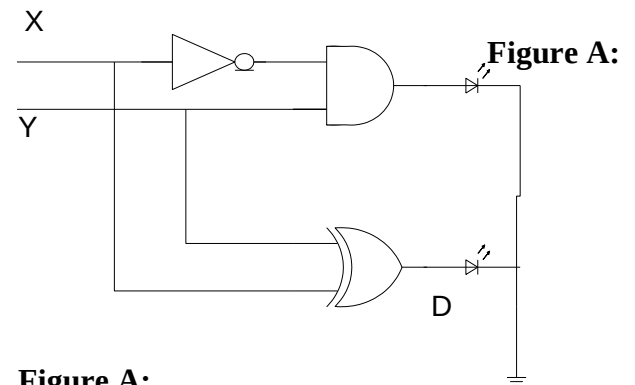
X	Y	B	D
0	0		
0	1		
1	0		
1	1		

Truth Table B:

Input		Output	
X	Y	B	D
0	0		
0	1		
1	0		
1	1		

Truth Table C:

X	Y	B	D
0	0		
0	1		
1	0		
1	1		



Test Problem:

1. What is the importance of R-1's complement in the computers system as a whole?
2. How do computers use the 1's complement in the arithmetic operation of a computer system?
3. Differentiate the 1's and 2's complement. Site each other advantages and disadvantages with regards to processing speed, complexity of the circuit, and reliability in arithmetic operations.
4. As a computer scientist, what complementing procedure do you recommend in computer systems?
5. Is it easier for computers to use complements in subtraction operation and would you recommend this process in the traditional manual process that we know since elementary years?
6. Design a Combinational Logic Circuit that will generate the 9's compliment of a BCD digit.

Experiment # 7A**Rating:** _____**Combinational Logic Circuit The Binary Full Adder Circuit****Objective:**

At the end the experiment, the students should be able to fabricate a binary full adder circuit having the same function with that of the Arithmetic Logic Unit (ALU) of the computer.

Materials:

IC# 7486 IC # 7408 IC # 7432 Connecting wires Light Emitting Diodes
5 Volts DC Power Supply Logic Probe Breadboard

Reminders:

Laboratory equipments & materials may cause harm to the researchers, please wear appropriate laboratory uniforms and protective gears for your safety. For ladies who have long hair and jewelries, it must be fixed to protect you from accidents **Theories:**

A full adder is a combinational circuit that forms the arithmetic sum of three input bits. Two of the input variables represent the two significant bits to be added. The third input variable represents the carry from the previous lower significant position. The outputs are necessary because the binary variables (X&Y) give the sum and carry (S & C respectively).

Procedures:

1.] Connect the circuit shown in the separate sheet.

*Make sure that the Power supply is set to **+5 VOLTS ONLY**

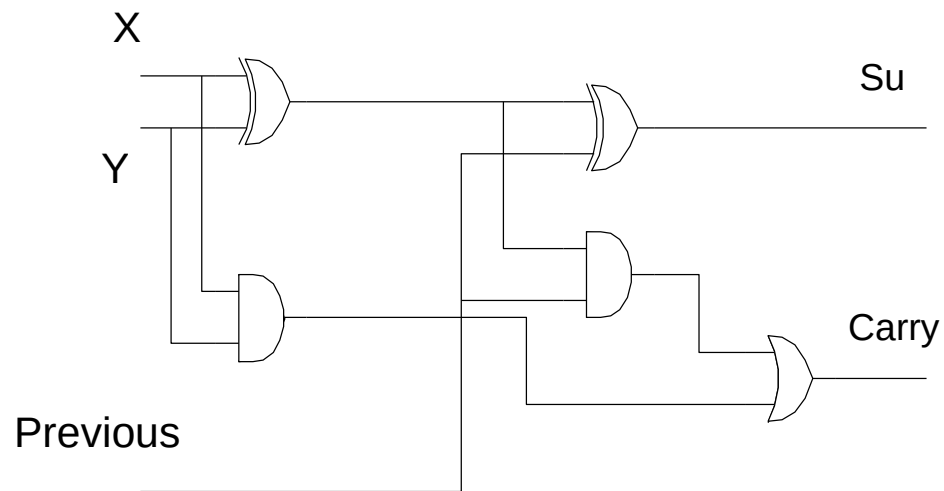
2.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

Output	Output Value
Led —OFF	0
Led ON	1

4.] Compare the circuit with the previous exercise (Half adder circuit).



Truth table:

X	Y	C	CARRY out	SUM
0	0	0		
0	0	1		
0	1	1		
0	1	0		
1	0	1		
1	0	0		
1	1	0		
1	1	1		

Conclusions/ Recommendations:

Test Problem:

1. Will it be possible to use the half-adder circuit and one's complement circuit be used as a full-subtractor circuit? Draw the circuit and prove the theory using Truth tables.
2. Can the half-adder circuit, Fig. A, B, C (PAGE 34), be replaced with a NAND or NOR gates? Draw the Logic Circuit diagram.
3. From problem #2, which of the following could process faster: Assuming each gate could process at a speed of 50 nSecs?
 - a. Fig. A, B, C of PAGE 34
 - b. Modified (Using NAND and NOR) Fig. A, B, C.
4. Can a Half-adder circuit be used to perform a full-adder circuit? Will it be as effective as the full-adder circuit? Draw the circuit and prove the theory using Truth tables.
5. Will it be possible to use the half-adder circuit and one's complement circuit to function as a half-subtractor circuit? Draw the circuit and prove the theory using Truth tables.

Experiment # 7B**Rating:** _____ [

Combinational Logic Circuit The Binary Full Subtractor Circuit

Objective:

At the end of the experiment, the student should be able to construct a Full subtractor circuit having the same function with that of the Arithmetic Logic Unit (ALU) of the computer.

Materials:

IC # 7486 IC # 7432 IC # 7408 Connecting wires Light Emitting Diode
 6 Volts DC Power Supply Logic Probe Breadboard

Reminders:

Laboratory equipments & materials may cause harm to the researchers, please wear appropriate laboratory uniforms and protective gears for your safety. For ladies who have long hair and jewelries, it must be fixed to protect you from accidents

Discussions:

A full subtractor is a combinational circuit that forms the arithmetic difference of three input bits. Two of the input variables represent the two significant bits to be subtracted. The third input variable represents the borrow from the previous lower significant position. The outputs are necessary because the binary variables (X&Y) give the difference and borrow (D & B, respectively).

Procedures:

1.] Connect the circuit shown in the separate sheet.

*Make sure that the Power supply is set to **+5 VOLTS ONLY**

2.] Input the values stated in the truth table.

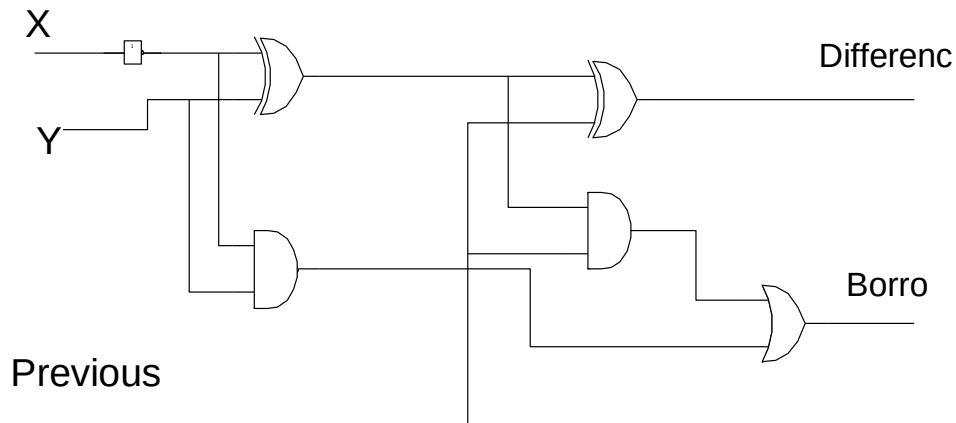
Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

Output	Output Value
--------	--------------

Led —OFF	0
Led ON	1

4.] Compare the circuit with the previous exercise (Half subtractor circuit).



Truth table:

X	Y	B	Borrow	Difference
0	0	0		
0	0	1		
0	1	1		
0	1	0		
1	0	1		
1	0	0		
1	1	0		
1	1	1		

Conclusions/ Recommendations:

Test Problem:

- Will it be possible to use the half-subtractor circuit and one's complement circuit be used as a full-subtractor circuit? Draw the circuit and prove the theory using Truth tables.

-
2. Can the half- subtractor circuit, Fig. A, B, C (PAGE 41), be replaced with a NAND or NOR gates? Draw the Logic Circuit diagram.
 3. From problem #2, which of the following could process faster: Assuming each gate could process at a speed of 50 nSecs?
 - a. Fig. A, B, C of PAGE 41
 - b. Modified (Using NAND and NOR) Fig. A, B, C.
 4. Can a Half- subtractor circuit be used to perform a full- subtractor circuit? Will it be as effective as the full– subtractor circuit? Draw the circuit and prove the theory using Truth tables.
 5. Will it be possible to use the half- subtractor circuit and one's complement circuit to function as a half-adder circuit? Draw the circuit and prove the theory using Truth tables.

Experiment # 7C**Rating:** _____**Combinational Logic Circuits The Binary Adder- Subtractor Circuit****Objective:**

At the end of the experiment, the students should be able to formulate a Adder/Subtractor circuit having the same function with that of the Arithmetic Logic Unit (ALU) of the computer.

Materials:

IC# 7483 IC # 7486 IC # 7432 IC# 7408 Light Emitting Diodes
5 Volts DC Power Supply Logic Probe Breadboard

Reminders:

Laboratory equipment's & materials may cause harm to the researchers, please wear appropriate laboratory uniforms and protective gears for your safety. For ladies who have long hair and jewelries, it must be fixed to protect you from accidents **Theories:**

A full adder- subtractor is a combinational circuit that forms the arithmetic sum or difference of three input bits. Two of the input variables represent the two significant bits to be added or subtracted. The third input variable represents the carry or borrows from the previous lower significant position. The outputs are necessary because the binary variables (X&Y) give the sum or difference and carry or borrow.

Procedures:

1.] Connect the circuit shown in the separate sheet.

*Make sure that the Power supply is set to **+5 VOLTS ONLY**

2.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

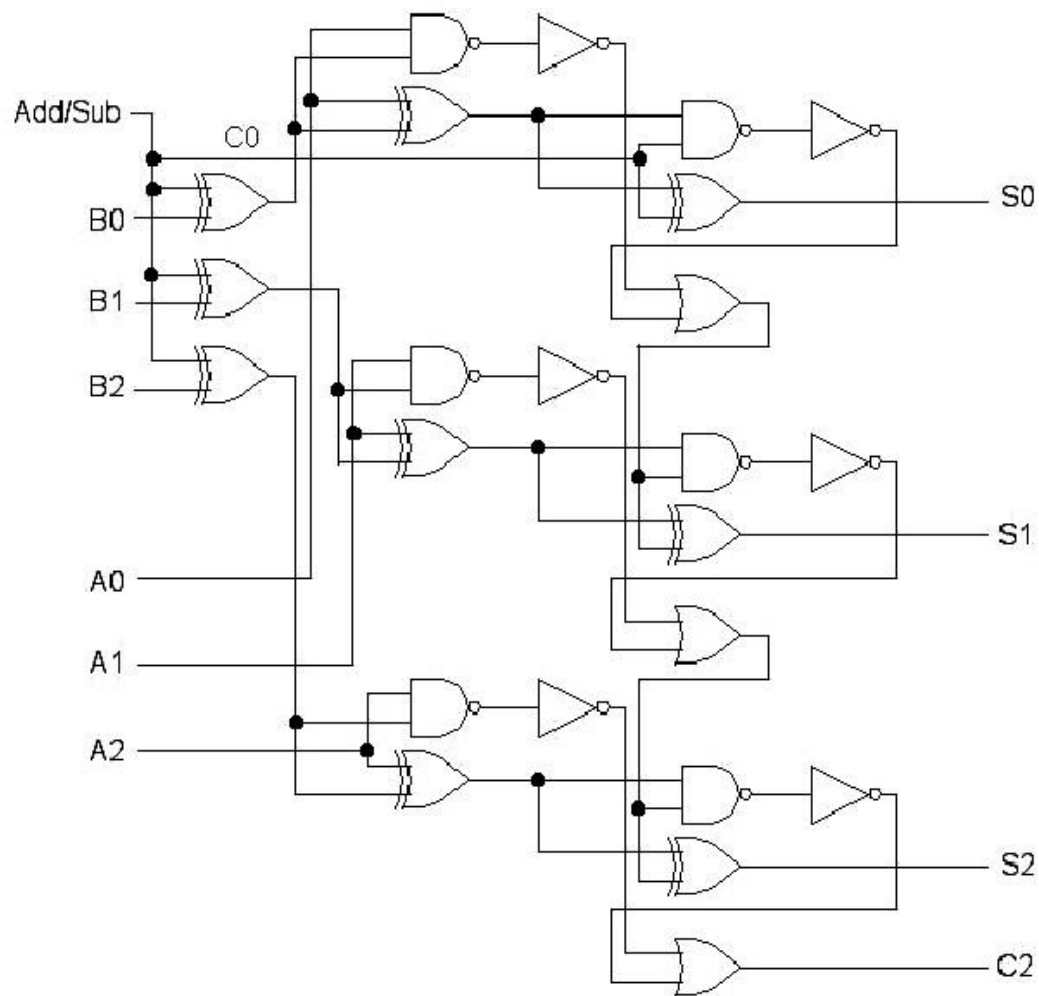
Output	Output Value
Led —OFF	0
Led ON	1

4.] Connect/ release the selector switch in the ground in order to add / subtract.

Operation Selection	Output Value
M = 0	addition
M = 1	Subtraction

5.] Repeat the steps #1 & #2.

6.] Compare this exercise with the previous exercises (half adder, half subtractor, Full adder or Full subtractor).



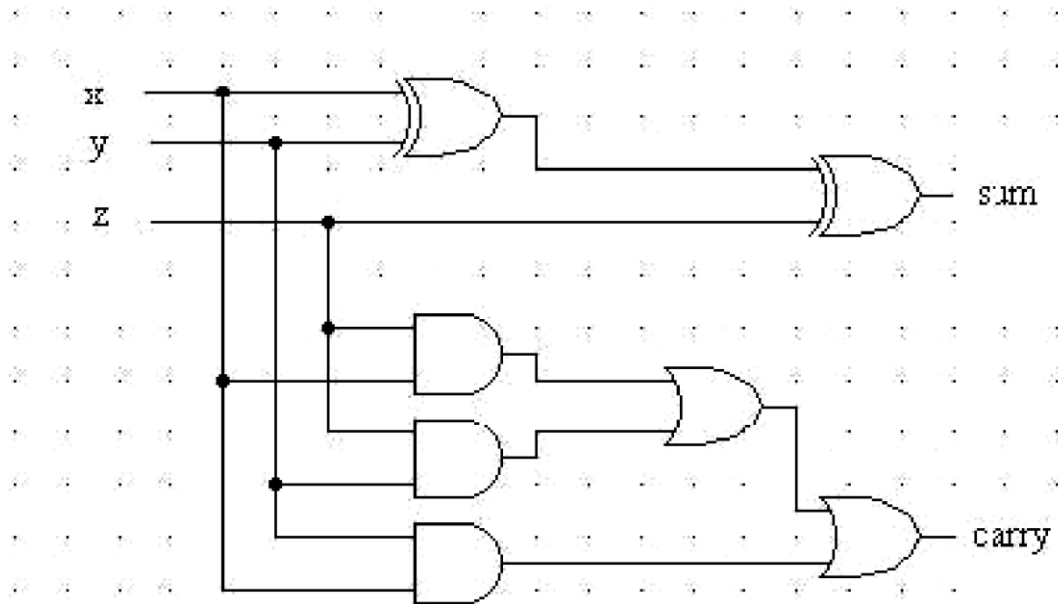
Truth table:

INPUT A			INPUT B			Output (M=0)				Output (M=1)			
A2	A1	A0	B2	B1	B0	C	S2	S1	S0	B	D2	D1	D0
0	0	0	0	0	0								
0	0	1	1	1	1								
0	1	0	1	0	0								
0	1	1	1	1	1								
1	0	0	1	1	0								
1	0	1	0	0	0								
1	1	0	0	0	1								
1	1	1	0	0	1								
0	0	0	1	1	0								
0	0	1	0	1	0								
0	1	0	0	0	1								
0	1	1	0	0	0								
1	0	0	0	1	0								
1	0	1	0	0	0								
1	1	0	0	0	0								
1	1	1	0	0	1								

Conclusions/ Recommendations:

Test Problem:

1. Using 4 exclusive-OR gates and a 4-bit full-adders MSI circuit, construct a 40bit parallel adder/Subtractor. Use an input select variable V so that when $V=0$, the circuit adds and $V=1$, the circuit subtracts. (Hint use 2's complement subtraction.)
2. In the circuit shown below, prove that it could really function as a full-adder circuit.



3. Create a VHDL program for the full adder / Subtractor circuit.

Experiment # 8A

Rating: _____

*Combinational Logic Circuits The Binary Multiplier Circuit***Objective:**

At the end of the experiment, the students should be able to formulate a multiplier circuit having the same function with that of the Arithmetic Logic Unit (ALU) of the computer. **Materials:**

IC# 7483 IC # 7486 IC # 7432 IC# 7408 Light Emitting Diodes
5 Volts DC Power Supply Logic Probe Breadboard

Reminders:

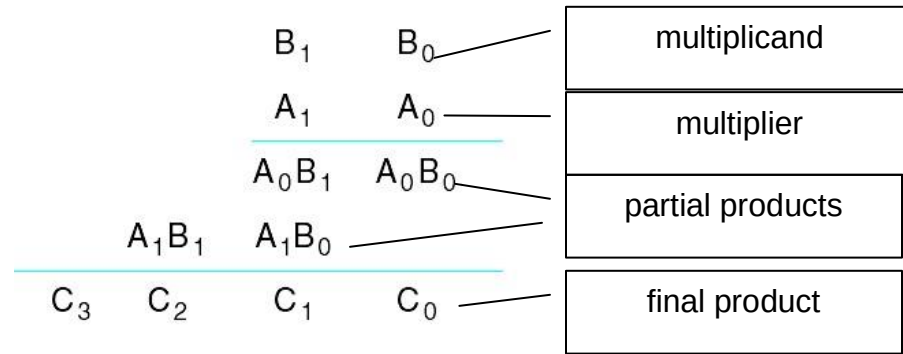
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Theories:

A binary multiplier is a electronic circuit used in digital electronics, such as a computer, to multiply two binary numbers. It is built using binary adders. The basic implementation of a Binary Multiplier is to:

- First **AND** each of the bits of the Multiplicand (defined as B0, B1, B2, for all the bits that you have) with the first bit of the multiplier A0). A0 **AND** B0 becomes the first bit of the output - C0.
- The second step is to **AND** each of the bits of the multiplicand with the next bit of the multiplier.
- The third step is adding these using a Binary Adder to the remaining bits from the previous group **AND** ({B0, B1, B2, ...} **AND** {A0} in this case). The least significant bit from the adder becomes C1, the next output bit.
- Repeat step 2 and 3 until all bits of the Multiplier are used. The carry from the last adder becomes the final bit in the product.

The multiplicand is multiplied by each bit of the multiplier, starting from the least significant bit. Each such multiplication forms a partial product. Successive partial products are shifted one bit to the left. The final product is obtained from the sum of the partial products.



Procedures:

1.] Connect the circuit shown in the above. Make sure that the Power supply is set to **+5 VOLTS ONLY**

2.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

Output	Output Value
Led —OFF	0
Led ON	1

Multiplicand				Multiplier			Product							

Conclusions/ Recommendations:

Experiment 8B

MUX/DEMUX USING 74153 & 74139

Aim: - To verify the truth table of MUX and DEMUX .

Apparatus Required: -

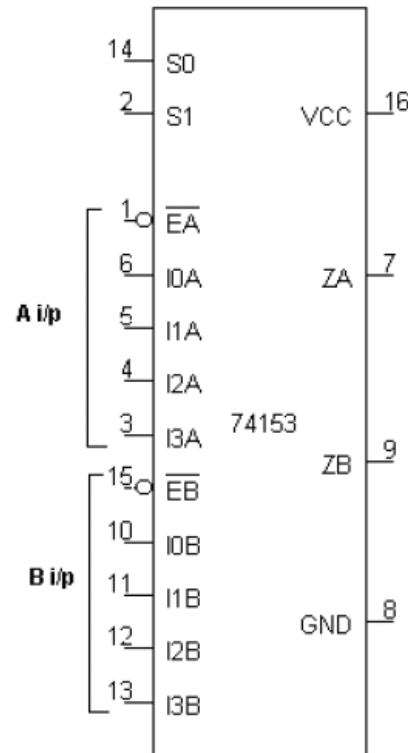
IC 74153, IC 74139 etc.

Procedure: - (IC 74153)

1. The Pin [16] is connected to + Vcc.
2. Pin [8] is connected to ground.
3. The inputs are applied either to 'A' input or 'B' input.
4. If MUX 'A' has to be initialized, Ea is made low and if MUX 'B' has to be initialized, Eb is made low.
5. Based on the selection lines one of the inputs will be selected at the output and thus verify the truth table

Procedure: - (IC 74139)

1. The inputs are applied to either 'a' input or 'b' input
2. The demux is activated by making Ea low and Eb low.
3. Verify the truth table .

Pin Details: -**Truth Table: -**

CHANNEL – A							
INPUTS					SELECT LINES		O/P
$\bar{E}_a$	I _{0a}	I _{1a}	I _{2a}	I _{3a}	S1	S2	Z _{a(v)}
1	X	X	X	X	X	X	0
0	0	X	X	X	0	0	0
0	1	X	X	X	0	0	1
0	X	0	X	X	0	1	0
0	X	1	X	X	0	1	1
0	X	X	0	X	1	0	0
0	X	X	1	X	1	0	1
0	X	X	X	0	1	1	0
0	X	X	X	1	1	1	1

CHANNEL – B							
INPUTS					SELECT LINES		O/P
$\bar{E}_a$	I _{0b}	I _{1b}	I _{2b}	I _{3b}	S1	S2	Z _{a(v)}
1	X	X	X	X	X	X	0
0	0	X	X	X	0	0	0
0	1	X	X	X	0	0	1
0	X	0	X	X	0	1	0
0	X	1	X	X	0	1	1
0	X	X	0	X	1	0	0
0	X	X	1	X	1	0	1
0	X	X	X	0	1	1	0
0	X	X	X	1	1	1	1

Pin Details: -Truth Table For Demux: -

CHANNEL – A							CHANNEL – B						
Inputs			Outputs				Inputs			Outputs			
$\bar{E}a$	$S1a$	$S0a$	$Y0a$	$Y1a$	$Y2a$	$Y3a$	$\bar{E}b$	$S1b$	$S0b$	$Y0b$	$Y1b$	$Y2b$	$Y3b$
1	X	X	1	1	1	1	1	X	X	1	1	1	1
0	0	0	0	1	1	1	0	0	0	0	1	1	1
0	0	1	1	0	1	1	0	0	1	1	0	1	1
0	1	0	1	1	0	1	0	1	0	1	1	0	1
0	1	1	1	1	1	0	0	1	1	1	1	1	0

Conclusion: -**Test Problem:**

1. Create a VHDL program for the multiplier circuit.
2. Redesign the 3X4 bits multiplier into a 4X4 bits multiplier

Experiment # 8C**Rating:** _____**Combinational Logic Circuit The Demultiplexer/Multiplexer circuit****Objective:** At the end of the experiment, the student should be able to:

- Devise a combinational logic circuit that will route the input to one of the many outputs using a switch input.

Materials:

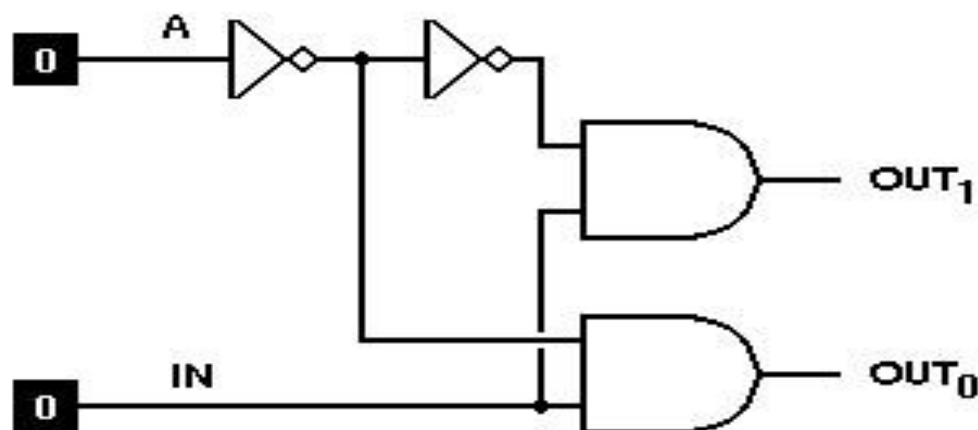
IC# 7404	IC # 7408	IC # 74138	Connecting wires	Light Emitting Diodes
Power Supply		Logic Probe	Breadboard	

Reminders:

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Discussions:

This circuit takes a single data input and one or more address inputs, and selects which of multiple outputs will receive the input signal. Using the address inputs as a binary number and producing an output signal on the single output that matches the binary address input can also use the same circuit as a decoder. In this application, the data input line functions as a circuit enabler — if the circuit is disabled, no output will show activity regardless of the binary input number. A one-line to two-line decoder/demultiplexer is shown below.



This circuit uses the same AND gates and the same addressing scheme as the two-input multiplexer circuit shown in these pages. The basic difference is that it is the inputs that are combined and the outputs that are separate. By making this change, we get a circuit that is the inverse of the two-input multiplexer.

If you were to construct both circuits on a single breadboard, connect the multiplexer output to the data IN of the demultiplexer, and drive the (A) address inputs of both circuits with the same signal, you would find that the initial X0 input would be transmitted to OUT0 and the X1 input would reach only OUT1. The one problem with this arrangement is that one of the two outputs will be inactive while the other is active.

To retain the output signal, we need to add a latch circuit that can follow the data signal while it's active, but will hold the last signal state while the other data signal is active. An excellent circuit for this is the D (or Data) Latch.

By placing a latch after each output and using the Addressing input (or its inverse) to control them, we can maintain both output signals at all times. If the Address input changes much more rapidly than the data inputs, the output signals will match the inputs faithfully.

Like multiplexers, demultiplexers are not limited to two data signals. If we use two addressing inputs, we can demultiplex up to four data signals. With three addressing inputs, we can demultiplex eight signals.

Procedure:

- 1.] Implement the figure shown below. Be sure to insert an LED in each output to be able to easily identify the result.
- 2.] Fill the table provided with the necessary results from the input values given.

*Make sure that the Power supply is set to **+5 VOLTS ONLY**

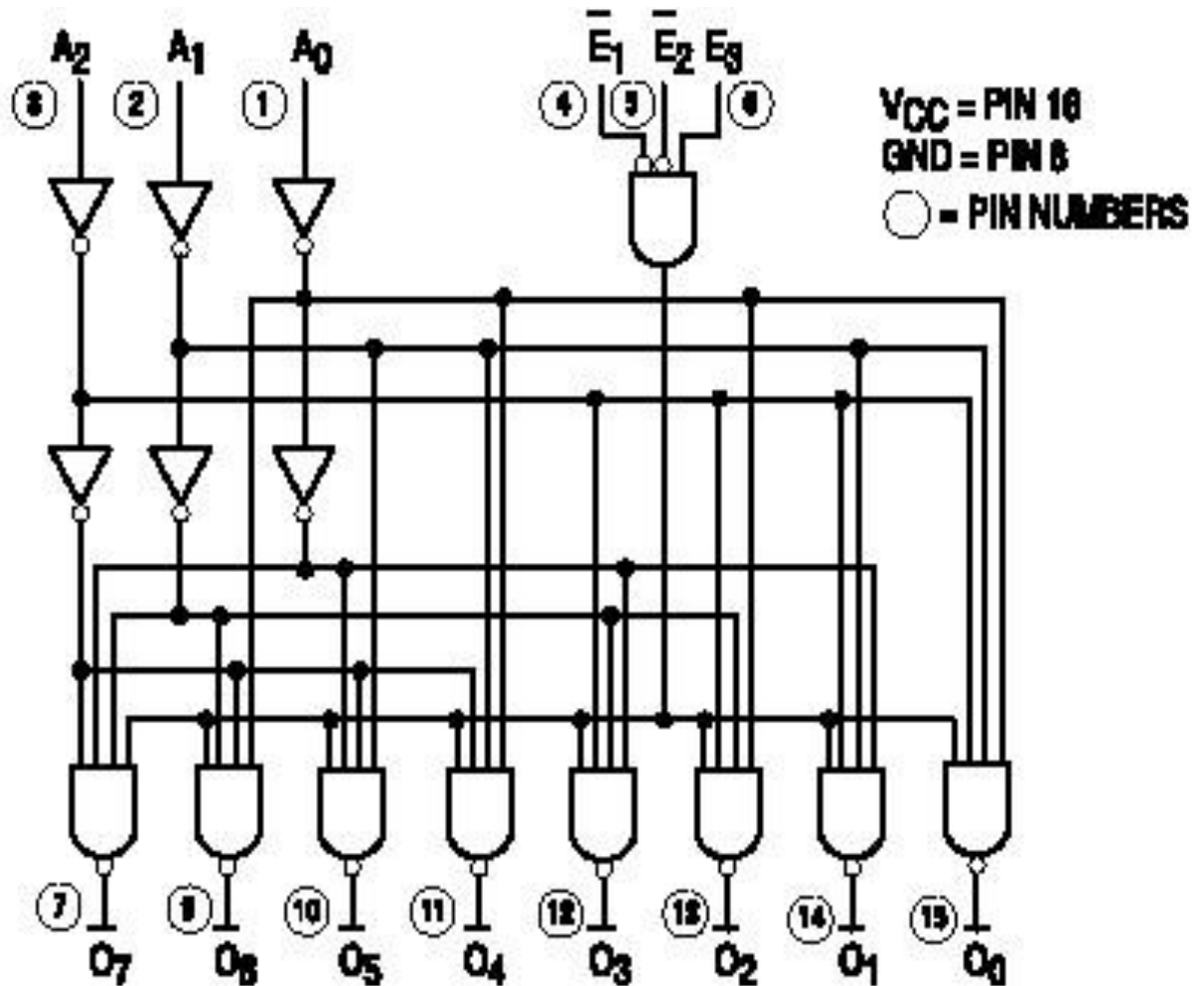
- 3.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

4.] Record the values given by the LEDs in the table below and compare the physical output with the theoretical output.

Output	Value Representation
Led —OFF	0
Led ON	1

Figure 1



This circuit is equivalent to IC # 74LS138 1 x 8 Decoder / Demultiplexer Circuit

Table 1:

Inputs							Outputs						
E1	E2	E3	A2	A1	A0	O ₀	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇
H	X	X	X	X	X	H	H	H	H	H	H	H	H
X	H	X	X	X	X	H	H	H	H	H	H	H	H
X	X	L	X	X	X	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	L	L	H	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	L	H	H	H	H	H	L	H	H	H	H
L	L	H	H	L	L	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	H	H	L	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

Note:

H (High) = 5 Volts	L (low) = GNDed	X (don't care) = 5 volts or GNDed
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Conclusions/ Recommendations:

Test Problem:

1. Implement a full-adder circuit with multiplexers.
2. A combinational circuit is defined by the following three functions:
 - a. $F1 = X'Y' + XYZ'$
 - b. $F2 = X' + Y$
 - c. $F3 = XY + X'Y'$

Design the circuit with a dual 4-line to 1-line multiplexer and an OR gate and Inverter.

3. Obtain an 8 X 1 multiplexer with a dual 4-line to 1-line multiplexers having separate enable inputs but common selection lines. Use a block diagram construction.
4. Create a VHDL program for the each problem.

Experiment # 9**Rating:** _____**Combinational Logic Circuits Seven segment decoder****Objective:** At the end of the experiment, the student should be able to:

- Design implement a decoder circuit, specifically a seven-segment decoder, and to familiarize the pin configuration of a dual 7-segment display
- Implement a decoder circuit, specifically a seven-segment decoder, and to familiarize the pin configuration of a dual 7-segment display **Materials:**

IC# 7447

IC # 7448

IC # 7408

Connecting Wires

Power Supply

Logic Probe

Breadboard

dual 7-segment display

Reminders:

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Theories:

The standard/common size of a decoder is n by 2^n wherein n is the number of circuit input. That means if the number of input signals is 3, the output will contain 8 unique signals. However, there are code detectors whose circuit set-up does not follow the expected output size. One of them is the 7447 IC that accepts 4 inputs and provides 10 unique output signals.

Another unusual feature of this IC is its dual functions. It is both a decoder and a driver. It is a BCD decoder. BCD stands for Binary Coded Decimal whose input bits start from 0000 to 1010. That is why the IC has only 10 outputs (0 - 9) and any signal exceeding the required input combinations may be considered **don't care** or unimportant signals, which hold no bearing on the resulting, output. Unlike the other decoders, each of 7447's output is composed of 7 signals.

With this feature, the result cannot be understood without another device to be attached to it. The device is called 7-segment display. Having this setup, the IC functions as a driver, since its output has sufficient current to light the corresponding segment/s in the 7-segment display. Each output combination will form 1 numeric/ decimal display represented by the segments that are active.

Procedures:

1.] Connect the circuit shown in figure A.

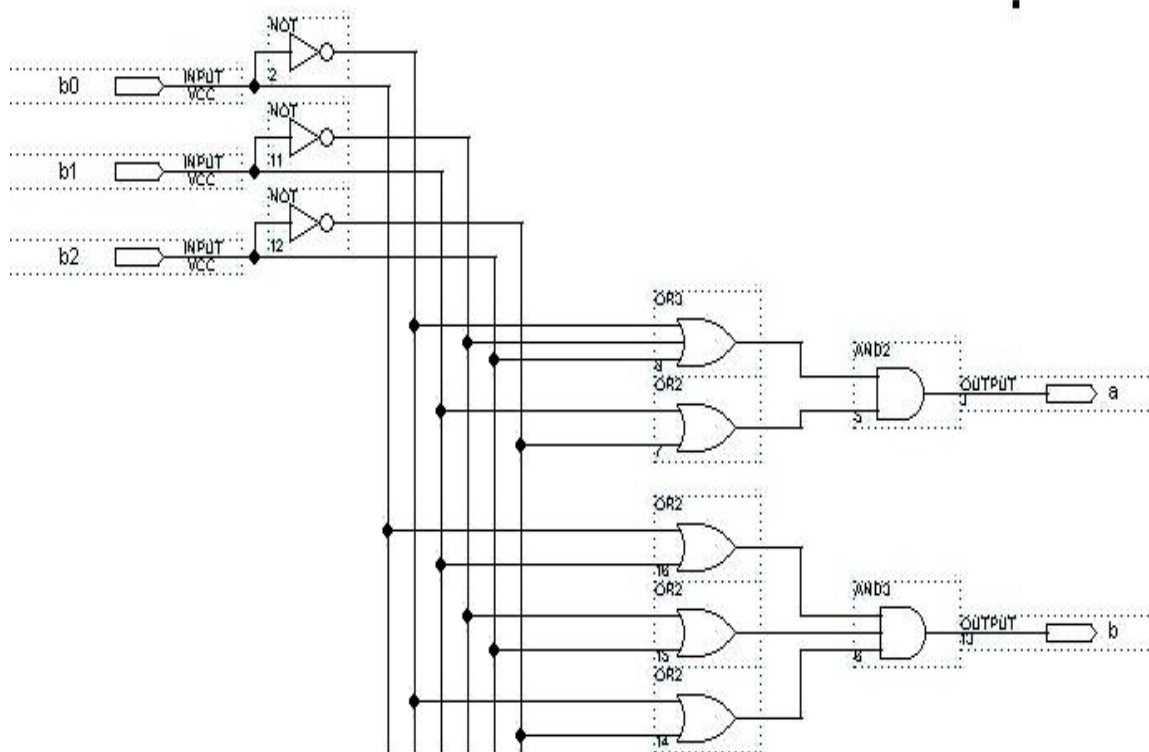
*Make sure that the Power supply is set to **+5 VOLTS ONLY**

2.] Input the values stated in the truth table.

Input Method	Output Value
Grounded (Negative supply)	0
Hang or connected to +5V	1

3.] Record the values given by the LEDs in the table below.

Output	Value Representation
Led —OFF	0
Led ON	1

Figure A**TRUTH TABLE:**

INPUT					OUTPUT ON THE SEVEN-SEGMENT DISPLAY						
A	B	C	D	#	a	b	c	d	e	f	g
0	0	0	0	0							
0	0	0	1	1							

0	0	1	0	2							
0	0	1	1	3							
0	1	0	0	4							
0	1	0	1	5							
0	1	1	0	6							
0	1	1	1	7							
1	0	0	0	8							
1	0	0	1	9							
1	0	1	0	10							
1	0	1	1	11							
1	1	0	0	12							
1	1	0	1	13							
1	1	1	0	14							
1	1	1	1	15							

Conclusions/ Recommendations:

Test Problem:

1. A combinational circuit is defined by the following three functions:

a. $F1 = X'Y' + XYZ'$

b. $F2 = X' + Y$

c. $F3 = XY + X'Y'$

Design the circuit with a Decoder and external gates.

2. A combinational circuit is defined by the following two functions:

a. $F1(X,Y) = \sum(0,3)$

b. $F2(X,Y) = \sum(1, 2, 3)$

Implement the Combinational Circuit by means of the Decoder and external NAND Gates.

3. Construct a 5 X 32 Decoder with four 3 X 8 decoder and a 2 X 4 decoder. Use a block diagram construction.
4. Draw the logic diagram of a 2-line to 4-line decoder using NOR gates only.
5. Create a VHDL program for the each problem.

Experiment # 10A

Rating: _____

Combinational Logic Circuits Magnitude Comparator

Objective:

At the end of the experiment, the students should be able to design a combinational logic circuit that will compare two values whether it is greater than, less than or equal.

Materials:

IC# 7404 IC # 7408 IC # 7432 IC # 7411 wires Power Supply Logic Probe Breadboard LEDs

Reminders:

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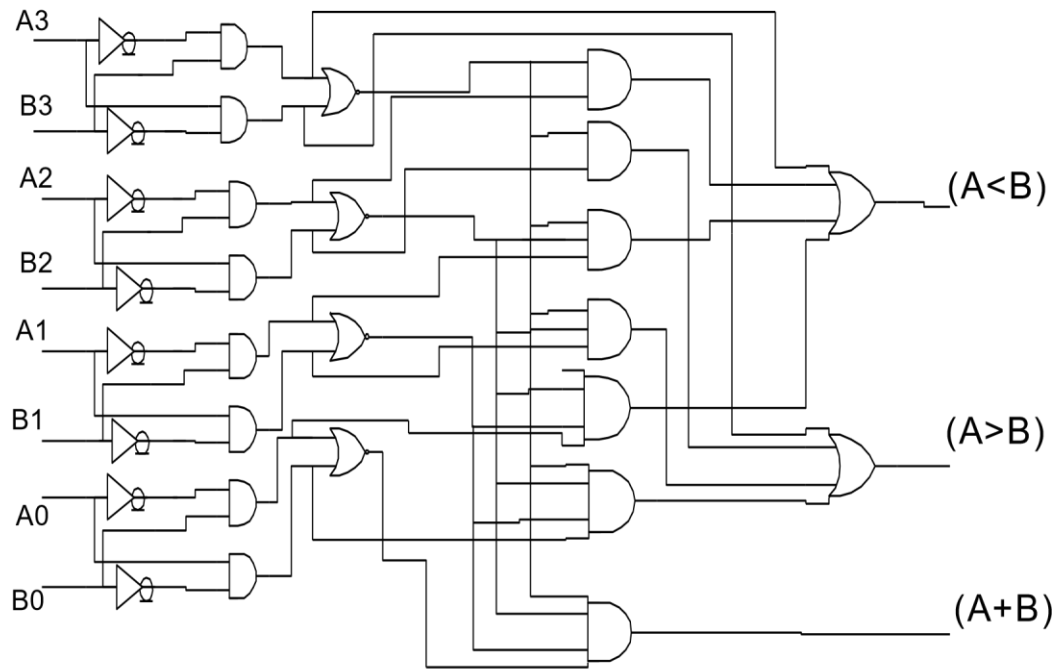
Theories:

Magnitude Comparator is circuit used in comparing two values. It can detect whether a number is greater than, less than or equal than the other. Example, if two values (A & B) are entered, it can compare whether A is greater than, less than or equal to B, or vice versa.

This type of circuit is one part of the Arithmetic Logic Unit (ALU) that performs the logic processing of the Central Processing Unit (CPU).

Procedures:

- 1.] Implement the figure shown below. Be sure to insert a LED in each output to be able to easily identify the result, whether A is Greater than, less than, or Equal to B.
- 2.] Fill the table provided with the necessary results from the input values given.



Input A				Input B				Output			
A3	A2	A1	A0	B3	B2	B1	B0		A > B	A < B	A = B
0	0	0	0	1	1	1	1				
0	1	1	1	1	0	0	0				
1	0	0	1	1	0	0	0				
0	0	1	1	0	0	1	1				
1	0	1	0	0	0	1	0				
0	0	0	0	1	0	0	0				
0	0	0	1	0	0	0	1				
1	1	0	1	1	1	1	0				
1	1	0	0	0	0	1	1				
0	0	0	0	0	0	0	0				
1	0	1	1	1	0	1	1				
1	1	1	1	1	1	1	0				
0	0	0	0	0	0	0	1				
1	0	1	0	1	0	0	1				
1	1	0	1	0	0	1	0				
0	1	0	0	1	0	1	0				
1	1	0	1	0	0	1	1				
0	1	0	0	1	0	1	1				
0	1	1	1	0	0	1	0				
0	1	1	0	1	0	0	0				
1	1	1	1	0	1	0	1				

Conclusions/ Recommendations:

Test Problem:

1. Design a circuit that compares two 4-bit numbers, A and B, to check if they are equal.
The circuit has one output X, so that $X = 1$ if $A = B$ and $X = 0$ if A is not equal to B.
2. Design a circuit that compares two 8-bit (Double digit) numbers, A and B, to check if they are equal, A is greater than B, and A is less than B . The circuit has three output X, $A > B$, $A < B$ & $A = B$.
3. Create a VHDL program for the each problem.

Experiment # 10B**Rating:** _____**4 BIT COMPARATORS**

Aim: - To verify the truth table of one bit and four bit comparators using logic Gates and IC 7485

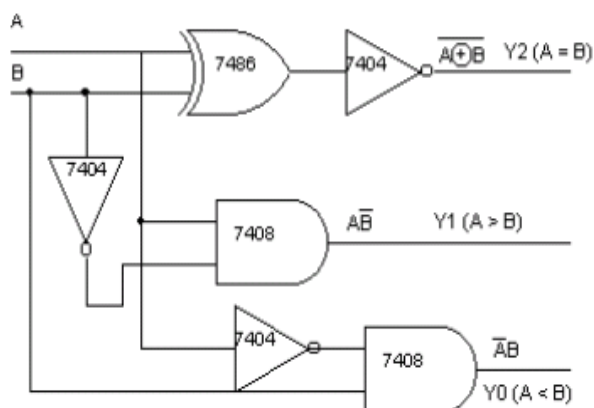
Apparatus Required: -

IC 7486, IC 7404, IC 7408, IC 7485etc.

Procedure: -

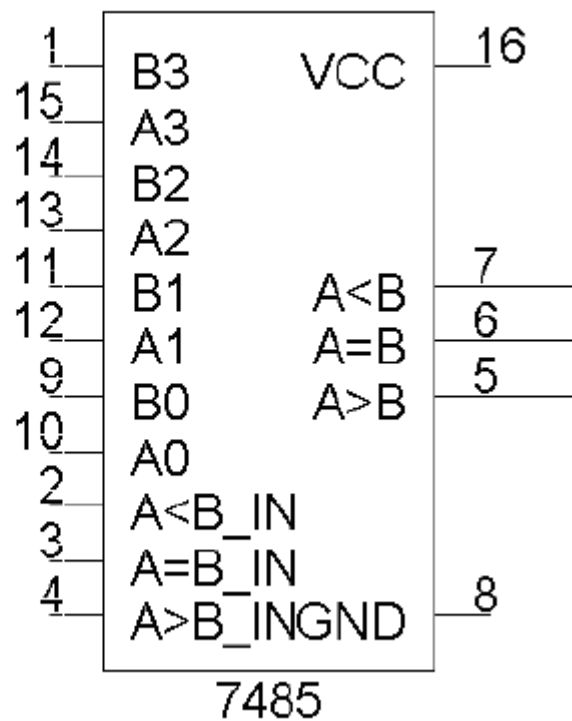
1. Verify the gates.
2. Make the connections as per the circuit diagram.
3. Switch on Vcc.
4. Applying i/p and Check for the outputs.
5. The readings of outputs should be tabulated .

Circuit diagram& truth tables:-

One Bit Comparator: -

A	B	Y1 (A>B)	Y2 (A = B)	Y3 (A < B)
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

4-bit Comparator



Tabular column :-

A ₃ B ₃	A ₂ B ₂	A ₁ B ₁	A ₀ B ₀	A>B	A=B	A<B	A>B	A=B	A<B
A ₃ >B ₃	X	X	X	X	X	X			
A ₃ <B ₃	X	X	X	X	X	X			
A ₃ =B ₃	A ₂ >B ₂	X	X	X	X	X			
A ₃ =B ₃	A ₂ <B ₂	X	X	X	X	X			
A ₃ =B ₃	A ₂ =B ₂	A ₁ >B ₁	X	X	X	X			
A ₃ =B ₃	A ₂ =B ₂	A ₁ <B ₁	X	X	X	X			
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ >B ₀	X	X	X			
A ₃ =B ₃	A ₂ =B ₂	A ₁ =B ₁	A ₀ <B ₀	X	X	X			

=====

Conclusion: -

Experiment # 11A**Rating:** _____**A BASIC, GATED, and CLOCKED LATCH****OBJECTIVE:**

At the end of the class, the student should be able to design a circuit that will show the output of a latch using combinational logic circuits.

MATERIALS:

IC # 7400	IC # 555	Resistor	IC # 7476
Wires	Capacitor	IC # 7404	LEDs
Potentiometer	IC # 74175	Power Supply	Bread Board

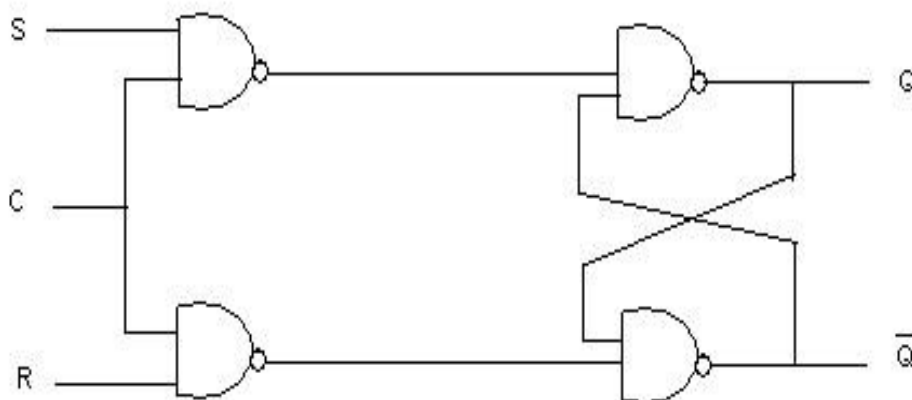
REMINDERS:

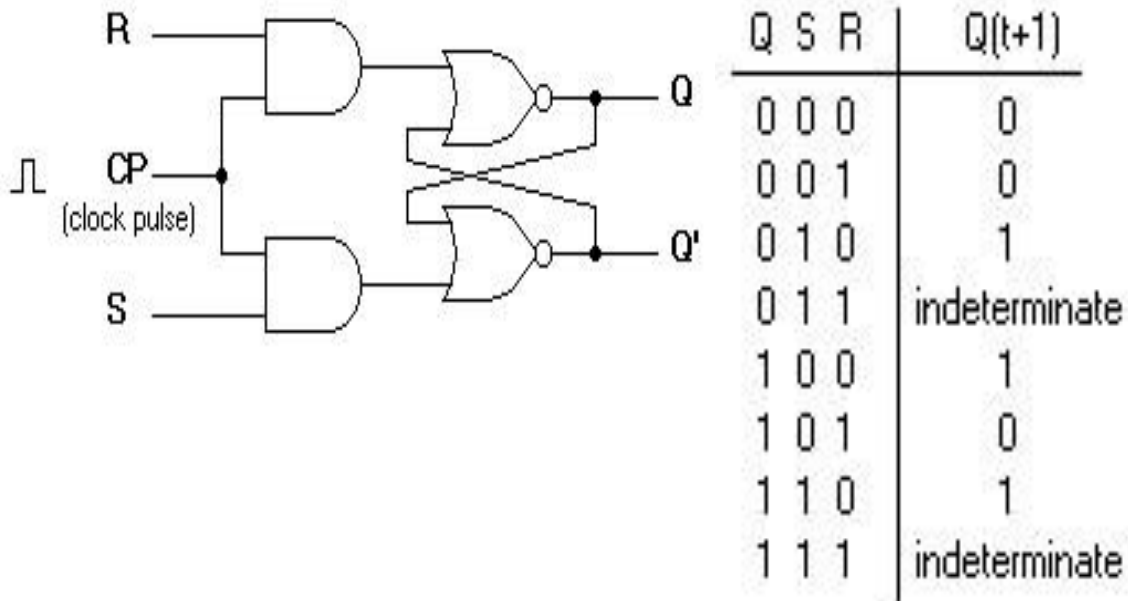
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SUMMARY:

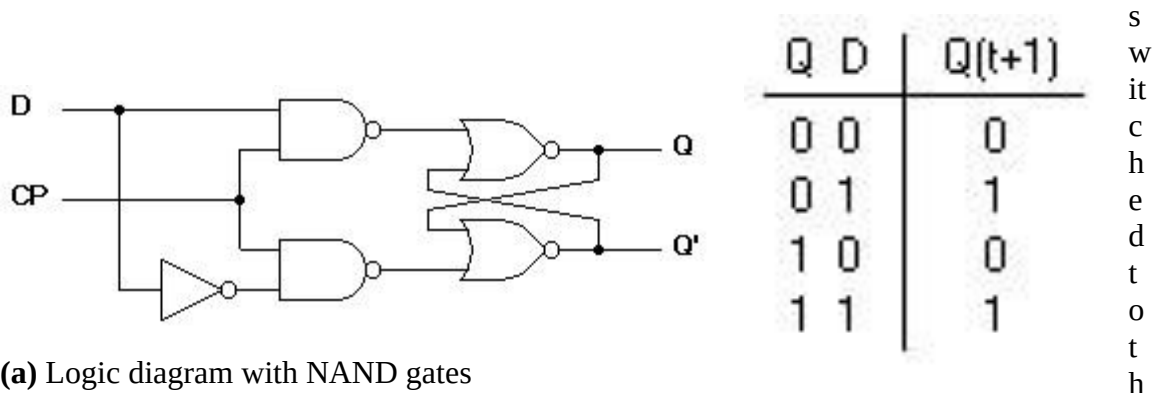
The clocked SR flip-flop shown consists of a basic NOR (sometimes NAND) flipflop and two AND gates. The outputs of the two AND gates remain at 0 as long as the clock pulse (or CP) is 0, regardless of the S and R input values.

When the clock pulse goes to 1, information from the S and R inputs passes through to the basic flip-flop. With both S=1 and R=1, the occurrence of a clock pulse causes both outputs to momentarily go to 0. When the pulse is removed, the state of the flip-flop is indeterminate, that is, either state may result, depending on whether the set or reset input of the flip-flop remains a 1 longer than the transition to 0 at the end of the pulse..

LOGIC DIAGRAMS:



The D flip-flop shown is a modification of the clocked SR flip-flop. The D input goes directly into the S input and the complement of the D input goes to the R input. The D input is sampled during the occurrence of a clock pulse. If it is 1, the flip-flop is



(a) Logic diagram with NAND gates

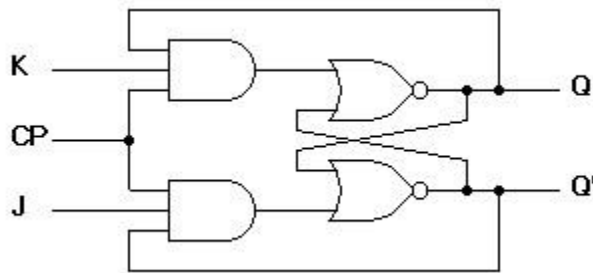
the set state (unless it was already set). If it is 0, the flip-flop switches to the clear state.

(b) Transition table

A JK flip-flop is a refinement of the SR flip-flop in that the indeterminate state of the SR type is defined in the JK type. Inputs J and K behave like inputs S and R to set and

clear the flip-flop (note that in a JK flip-flop, the letter J is for set and the letter K is for clear). When logic 1 inputs are applied to both J and K simultaneously, the flip-flop switches to its complement state, ie., if $Q=1$, it switches to $Q=0$ and vice versa.

A clocked JK flip-flop is shown. Output Q is ANDed with K and CP inputs so that the flip-flop is cleared during a clock pulse only if Q was previously 1. Similarly, output Q' is ANDed with J and CP inputs so that the flip-flop is set with a clock pulse only if Q' was previously 1.



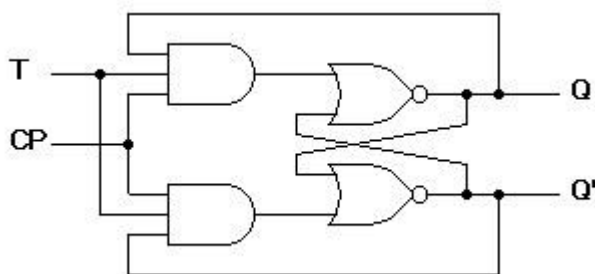
Q	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

(a) Logic diagram

(b) Transition table

Note that because of the feedback connection in the JK flip-flop, a CP signal which remains a 1 (while $J=K=1$) after the outputs have been complemented once will cause repeated and continuous transitions of the outputs.

To avoid this, the clock pulses must have a time duration less than the propagation delay through the flip-flop. The restriction on the pulse width can be eliminated with a master-slave or edge-triggered construction. The same reasoning also applies to the T flip-flop presented next.



(a) Logic diagram

(b) Transition table

The T flip-flop is a single input version of the JK flip-flop. As shown, the T flipflop is obtained from the JK type if both inputs are tied together. The output of the T flipflop "toggles" with each clock pulse.

The state of a flip-flop is changed by a momentary change in the input signal. This change is called a trigger and the transition it causes is said to trigger the flip-flop. The basic circuits require an input trigger defined by a change in signal level.

This level must be returned to its initial level before a second trigger is applied. Clocked flip-flops are triggered by pulses.

The feedback path between the combinational circuit and memory elements can produce instability if the outputs of the memory elements (flip-flops) are changing while the outputs of the combinational circuit that go to the flip-flop inputs are being sampled by the clock pulse.

A way to solve the feedback timing problem is to make the flip-flop sensitive to the pulse transition rather than the pulse duration.

The clock pulse goes through two signal transitions: from 0 to 1 and the return from 1 to 0. As shown, the positive transition is defined as the positive edge and the negative transition as the negative edge.

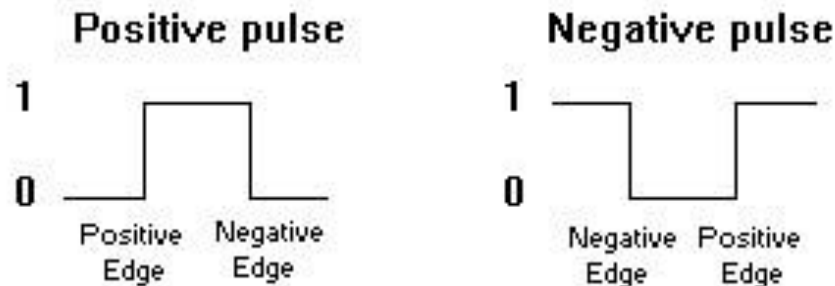


Figure: Definition of clock pulse transition

The clocked flip-flops already introduced are triggered during the positive edge of the pulse, and the state transition starts as soon as the pulse reaches the logic-1 level. If the other inputs change while the clock is still 1, a new output state may occur.

If the flip-flop is made to respond to the positive (or negative) edge transition only, instead of the entire pulse duration, then the multiple-transition problem can be eliminated.

Procedure:

1. Implement the figures (RS-, JK-, T-, and D-Flip flop) shown using NAND gates. Be sure to connect all Vcc's and GND's to the correct source line.
2. Display the outputs using the Light Emitting Diodes (LED).
3. Make sure to record the outputs using truth tables.
4. Repeat steps 1- 3 using NOR gates.
5. Prove and disprove the theories stated in the discussion

Conclusions/ Recommendations:

Problem:

1. What were the similarities noticed from the different Flip-Flops experimented that we can apply to our Memory Element Circuits?

Experiment # 11B

Rating: _____

FLIP-FLOP

Aim:- Truth table verification of Flip-Flops: (i) RS-Type
(ii) D- Type
(iii) T- Type.
(iv) JK-Type

Apparatus Required: -

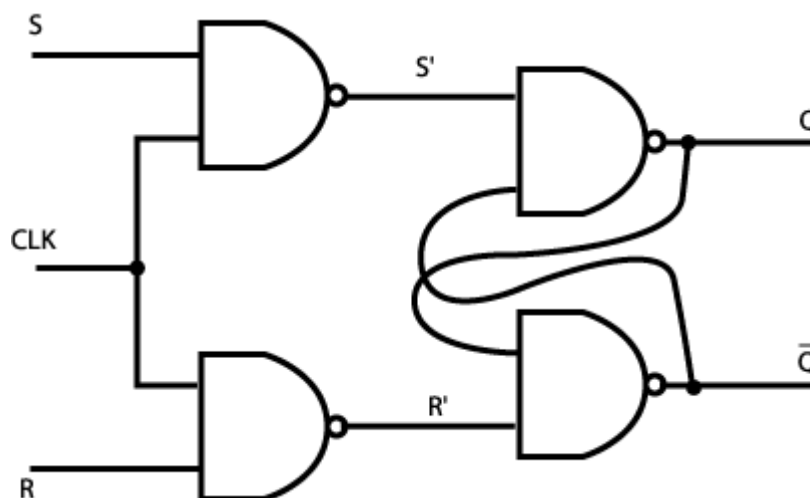
IC 7400, IC 7404 etc.

Procedure: -

1. Connections are made as per circuit diagram.
2. Verify the truth table for various combinations of inputs.

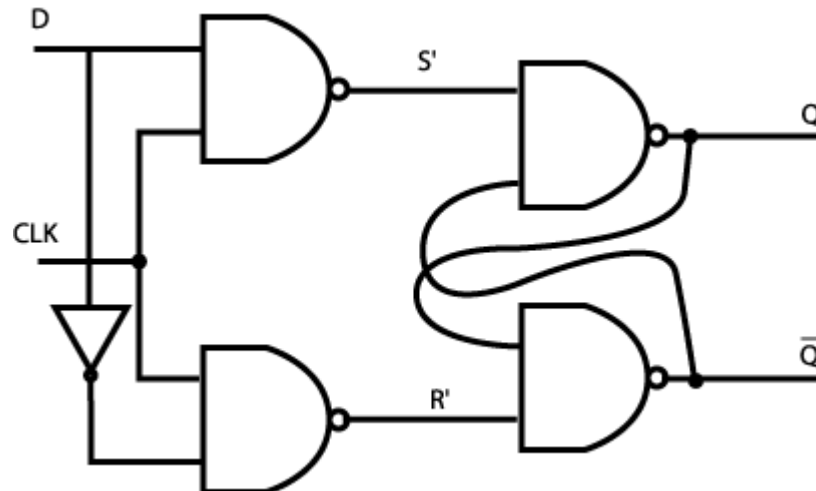
Circuit Diagram& Truth table:

i) RS Flip-Flop



$Q_n / \bar{Q}_n$	R	S	$Q_{n+1} / \bar{Q}_{n+1}$
0 1	0	0	
1 0	0	0	
0 1	0	1	
1 0	0	1	
0 1	1	0	
1 0	1	0	
0 1	1	1	
1 0	1	1	

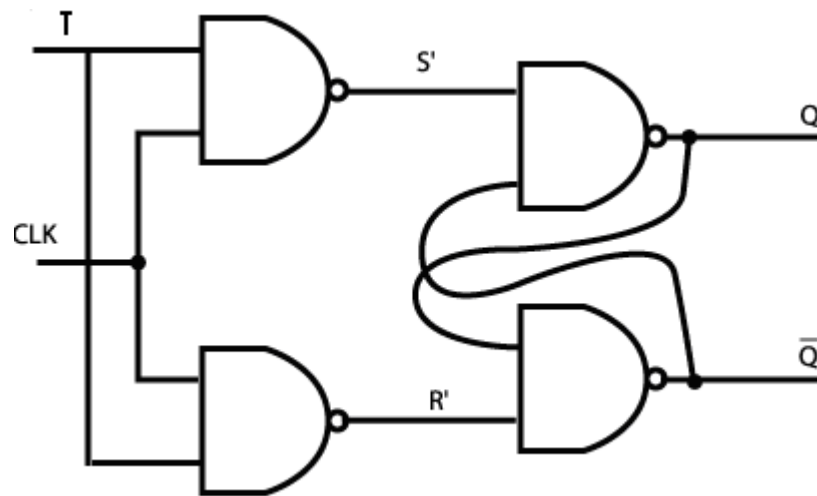
ii) D Flip-Flop



D Flip-Flop:-

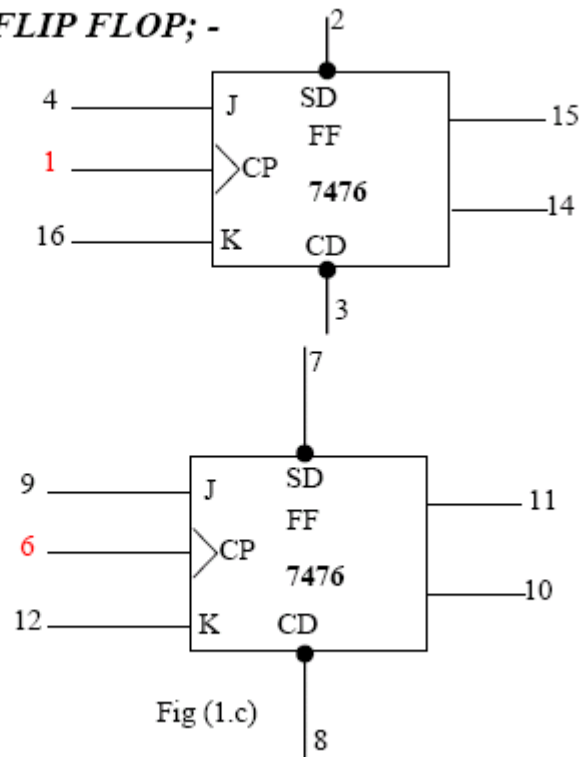
Preset	Clear	D	Clock	Q_{n+1}	$\overline{Q_{n+1}}$
1	1	0	$\neg$	0	1
1	1	1	$\neg$	1	0

iii) T Flip-flop



T Flip-Flop:-

Preset	Clear	T	Clock	Q_{n+1}	$\overline{Q_{n+1}}$
1	1	0	CLK	Q_n	$\overline{Q_n}$
1	1	1	CLK	$\overline{Q_n}$	Q_n

IC 7476 M/S JK FLIP FLOP; -**TRUTH TABLE FOR JK-FLIP FLOP (IC 7476); -**

SD Preset	CD Clear	Clock	J	K	OUTPUTS	
					Q	$\bar{Q}$
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H*	H*
H	H	↓	L	L	Q_0	$\bar{Q}_0$
H	H	↓	H	L	H	L
H	H	↓	L	H	L	H
H	H	↓	H	H	TOGGLE	
H	H	H	X	X	Q_0	$\bar{Q}_0$

*Unstable condition. It will not remain after C_n and P_n inputs return to their inactive (high) state

Conclusion:-

Experiment # 12**Rating:** _____**RANDOM ACCESS MEMORY(RAM)**

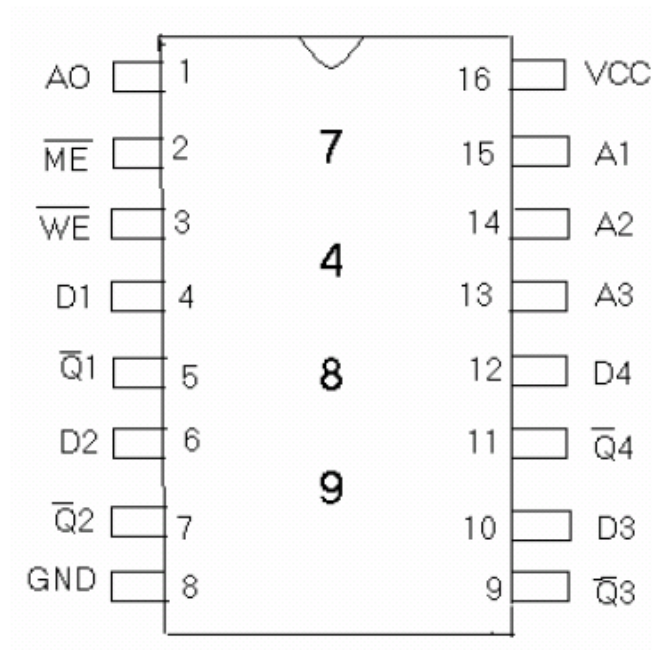
Aim: - To conduct an experiment to store a set of data in a RAM using IC 7489 starting from location ----- to location----- and retrieve the same data.

Apparatus Required: -

IC 7489, etc.

Procedure: -

1. Circuits connections are made to the appropriate pins of IC 7489
2. First you have to write the data and then read the data, for writing data make WE to low and ME input to low
3. For a 4-bit data select any address input from A0 to A9. for ex, select A3 to A0 and connect the data inputs/ outputs i.e., I/O4 – I/O1
4. Write a 4-bit data of your choice in each of the required address inputs or memory locations
5. By doing the above steps 2, 3 and 4 the data will be stored in the memory location
6. For reading data
 - a. make WE to high and ME input to low
 - b. disconnect the data inputs I/O4 – I/O1 from input lines and connect them to output lines to read the data
 - c. and then give the address inputs of the data you have stored and observe the outputs through I/O4 – I/O1.

Pin Diagram: -Operation:-

- RAM IC 7489 is 16 words x 4-bit Read/WriteMemory.
- The Truth Table for the RAM IC 7489 is given below.

Memory Enable	Write Enable	Operation
H	X	All data outputs are high
L	H	Read mode (data outputs are compliment of the RAM content).
L	L	Write mode (data inputs are written on to the memory; dataoutputs are compliment of the RAM content).

Example for Data Input:-

Address Inputs			
A3	A2	A1	A0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1

Data Inputs			
I/O4	I/O3	I/O2	I/O1
0	0	1	0
0	1	0	0
0	1	0	1
0	1	1	0

Example for Data Output:-

Address Inputs			
A3	A2	A1	A0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1

Data Outputs			
I/O4	I/O3	I/O2	I/O1
0	0	1	0
0	1	0	0
0	1	0	1
0	1	1	0

Conclusion:-

Rating: _____

Experiment # 13**SHIFT REGISTER**

Aim:- To study shift register using IC 7495 in all its modes i.e. SIPO/SISO, PISO/PIPO.

Apparatus Required: -

IC 7495, etc.

Procedure:-**Serial In Parallel Out(SIPO):-**

1. Connections are made as per circuit diagram.
2. Apply the data at serial i/p
3. Apply one clock pulse at clock 1 (Right Shift) observe this data at QA.
4. Apply the next data at serial i/p.
5. Apply one clock pulse at clock 2, observe that the data on QA will shift to QB and the new data applied will appear at QA.
6. Repeat steps 2 and 3 till all the 4 bits data are entered one by one into the shift register.

Serial In Serial Out(SISO):-

1. Connections are made as per circuit diagram.
2. Load the shift register with 4 bits of data one by one serially.
3. At the end of 4th clock pulse the first data 'd0' appears at QD.
4. Apply another clock pulse; the second data 'd1' appears at QD.

5. Apply another clock pulse; the third data appears at QD.
6. Application of next clock pulse will enable the 4th data 'd3' to appear at QD. Thus the data applied serially at the input comes out serially at QD

Parallel In Serial Out (PISO):-

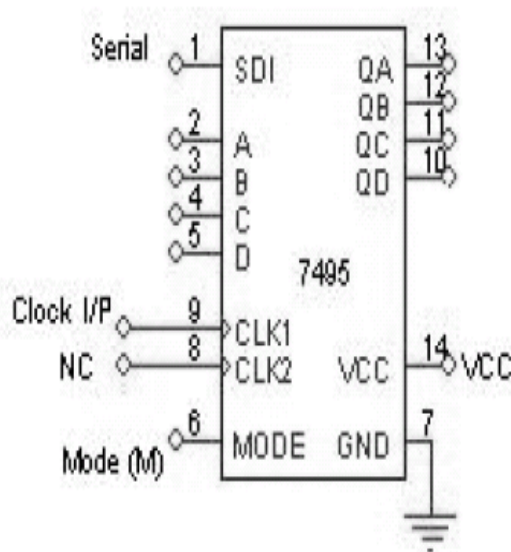
1. Connections are made as per circuit diagram.
2. Apply the desired 4 bit data at A, B, C and D.
3. Keeping the mode control M=1 apply one clock pulse. The data applied at A, B, C and D will appear at QA, QB, QC and QD respectively.
4. Now mode control M=0. Apply clock pulses one by one and observe the Data coming out serially at QD

Parallel In Parallel Out (PIPO):-

1. Connections are made as per circuit diagram.
2. Apply the 4 bit data at A, B, C and D.
3. Apply one clock pulse at Clock 2 (Note: Mode control M=1).
4. The 4 bit data at A, B, C and D appears at QA, QB, QC and QD respectively.

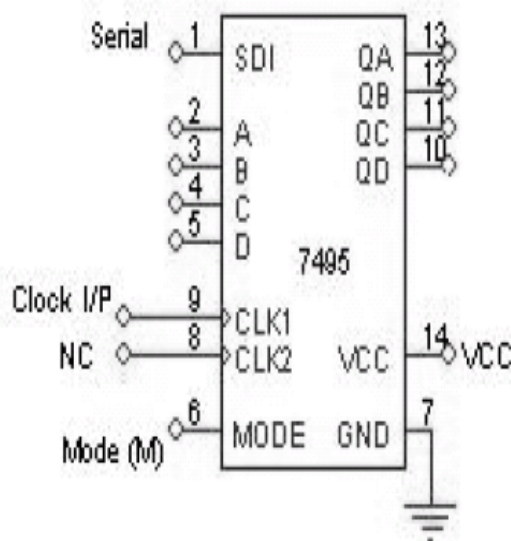
Circuit diagram :-

SIPO (Right Shift):-

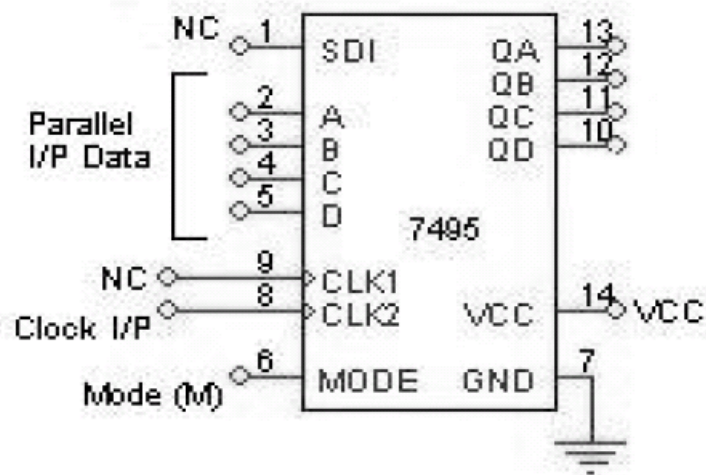


Clock	Serial i/p	QA	QB	QC	QD
1	0	0	X	X	X
2	1	1	0	X	X
3	1	1	1	0	X
4	1	1	1	1	0

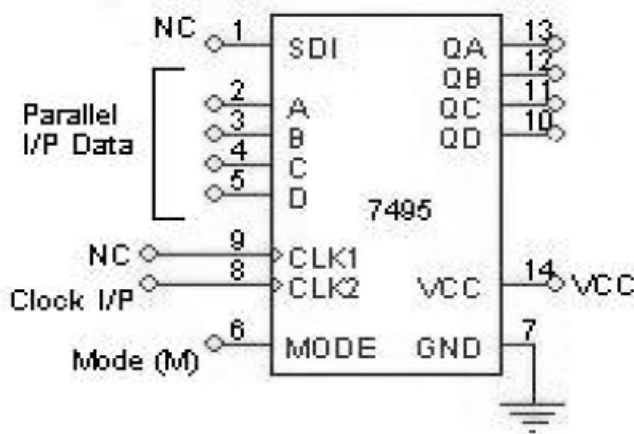
SISO:-



Clock	Serial i/p	QA	QB	QC	QD
1	do=0	0	X	X	X
2	d1=1	1	0	X	X
3	d2=1	1	1	0	X
4	d3=1	1	1	1	0=do
5	X	X	1	1	1=d1
6	X	X	X	1	1=d2
7	X	X	X	X	1=d3

PISO:-

Mode	Clock	Parallel i/p				Parallel o/p			
		A	B	C	D	QA	QB	QC	QD
1	1	1	0	1	1	1	0	1	1
0	2	X	X	X	X	X	1	0	1
0	3	X	X	X	X	X	X	1	0
0	4	X	X	X	X	X	X	X	1

PIPO:-

Clock	Parallel i/p				Parallel o/p			
	A	B	C	D	QA	QB	QC	QD
1	1	0	1	1	1	0	1	1

Conclusion:-

Experiment # 14

Rating: _____

ARITHMETIC LOGIC UNIT

Aim: - To verify the Function table of 4 bit ALU(IC 74181.

Apparatus Required: -

IC 74181, etc.

Procedure: -

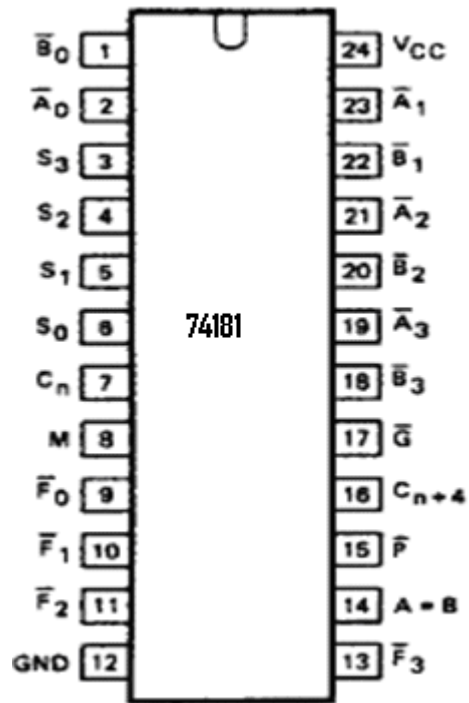
1. Connections are made as shown in the Circuit diagram.
2. Change the values of the inputs and verify at least 5 functions given in the function table.

Pin detail & Function table:-

PIN DESCRIPTION

PIN NO.	SYMBOL	NAME AND FUNCTION
1, 22, 20, 18	$\overline{B}_0$ to $\overline{B}_3$	operand inputs (active LOW)
2, 23, 21, 19	$\overline{A}_0$ to $\overline{A}_3$	operand inputs (active LOW)
6, 5, 4, 3	S_0 to S_3	select inputs
7	C_n	carry input
8	M	mode control input
9, 10, 11, 13	$\overline{F}_0$ to $\overline{F}_3$	function outputs (active LOW)
12	GND	ground (0 V)
14	A=B	comparator output
15	$\overline{P}$	carry propagate output (active LOW)
16	C_{n+4}	carry output
17	$\overline{G}$	carry generate output (active LOW)
24	V_{CC}	positive supply voltage

IC 74181



FUNCTION TABLES

MODE SELECT INPUTS				ACTIVE HIGH INPUTS AND OUTPUTS	
S_3	S_2	S_1	S_0	LOGIC (M=H)	ARITHMETIC ⁽²⁾ (M=L; C_n =H)
L	L	L	L	$\overline{A}$	A
L	L	L	H	$\overline{A+B}$	A + B
L	L	H	L	$\overline{AB}$	A + $\overline{B}$
L	L	H	H	logical 0	minus 1
L	H	L	L	$\overline{AB}$	A plus $\overline{AB}$
L	H	L	H	$\overline{B}$	(A + B) plus $\overline{AB}$
L	H	H	L	$A \oplus B$	A minus B minus 1
L	H	H	H	$\overline{AB}$	$\overline{AB}$ minus 1
H	L	L	L	$\overline{A+B}$	A plus AB
H	L	L	H	$\overline{A \oplus B}$	A plus B
H	L	H	L	B	(A + $\overline{B}$) plus AB
H	L	H	H	AB	AB minus 1
H	H	L	L	logical 1	A plus A ⁽¹⁾
H	H	L	H	$A + \overline{B}$	(A + B) plus A
H	H	H	L	A + B	(A + $\overline{B}$) plus A
H	H	H	H	A	A minus 1

Notes to the function tables

- Each bit is shifted to the next more significant position.
- Arithmetic operations expressed in 2s complement notation.

H = HIGH voltage level

L = LOW voltage level

MODE SELECT INPUTS				ACTIVE LOW INPUTS AND OUTPUTS	
S_3	S_2	S_1	S_0	LOGIC (M=H)	ARITHMETIC ⁽²⁾ (M=L; C_n =L)
L	L	L	L	$\overline{A}$	A minus 1
L	L	L	H	$\overline{AB}$	AB minus 1
L	L	H	L	$\overline{A+B}$	$\overline{AB}$ minus 1
L	L	H	H	logical 1	minus 1
L	H	L	L	$\overline{A+B}$	A plus (A + $\overline{B}$)
L	H	L	H	$\overline{B}$	AB plus (A + $\overline{B}$)
L	H	H	L	$\overline{A \oplus B}$	A minus B minus 1
L	H	H	H	$A + \overline{B}$	A + $\overline{B}$
H	L	L	L	$\overline{AB}$	A plus (A + B)
H	L	L	H	$A \oplus B$	A plus B
H	L	H	L	B	$\overline{AB}$ plus (A + B)
H	L	H	H	A + B	A + B
H	H	L	L	logical 0	A plus A ⁽¹⁾
H	H	L	H	$\overline{AB}$	AB plus A
H	H	H	L	AB	$\overline{AB}$ plus A
H	H	H	H	A	A

Notes to the function tables

- Each bit is shifted to the next more significant position.
- Arithmetic operations expressed in 2s complement notation.

H = HIGH voltage level

L = LOW voltage level

Experiment # 15**Rating:** _____**ASYNCHRONOUS COUNTER DESIGN AND MOD-N COUNTER**

Aim: - Realization of 3-bit Asynchronous counter and Mod-N counter design .

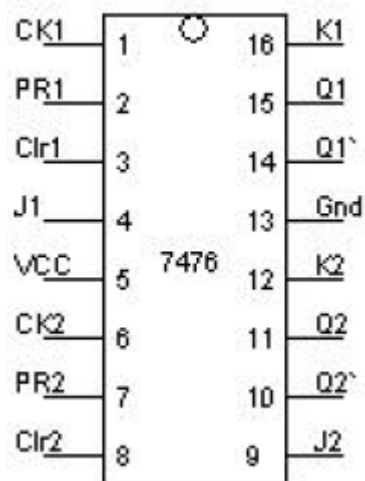
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Apparatus Required: -

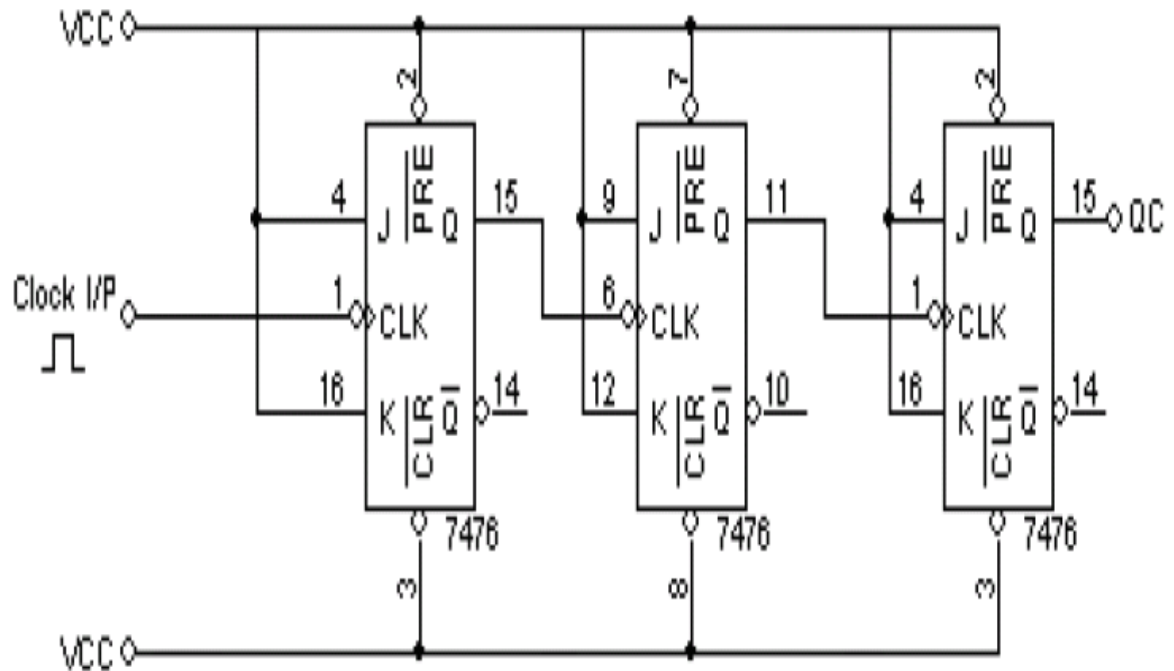
IC 7408, IC 7476, IC 7400, IC 7432 etc.

Procedure: -

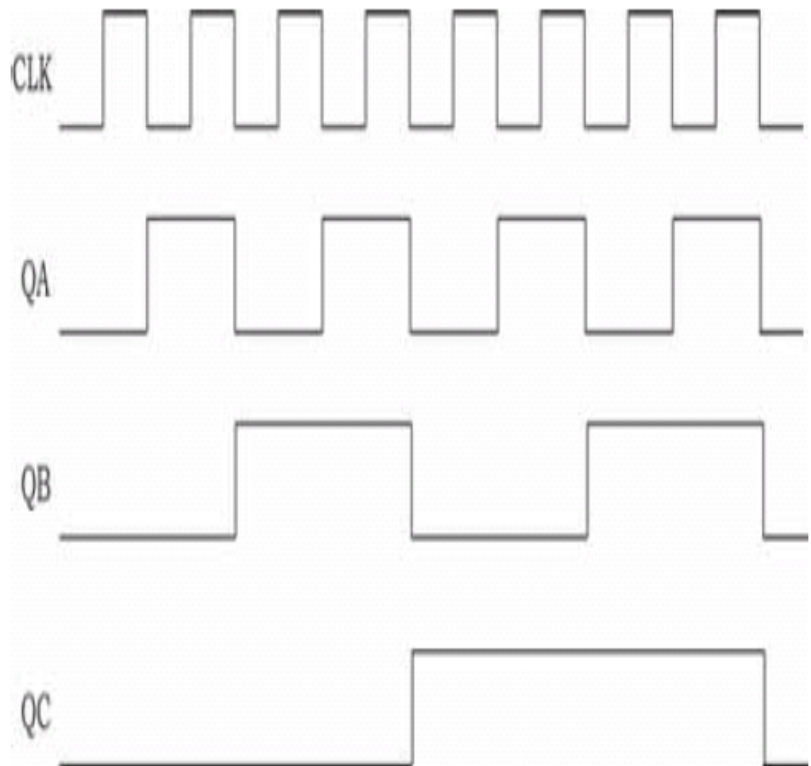
1. Connections are made as per circuit diagram.
2. Clock pulses are applied one by one at the clock I/P and the O/P is observed at QA, QB & QC for IC 7476.
3. Verify the Truth table .

Circuit Diagram:**Pin Details: -**

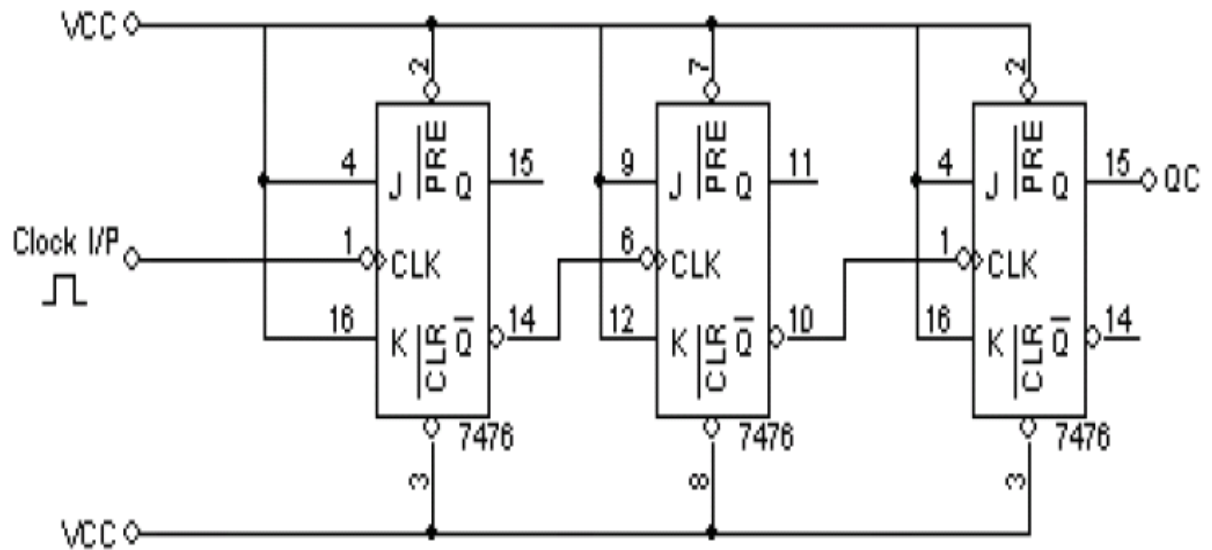
Circuit Diagram: - 3-Bit Asynchronous Up Counter



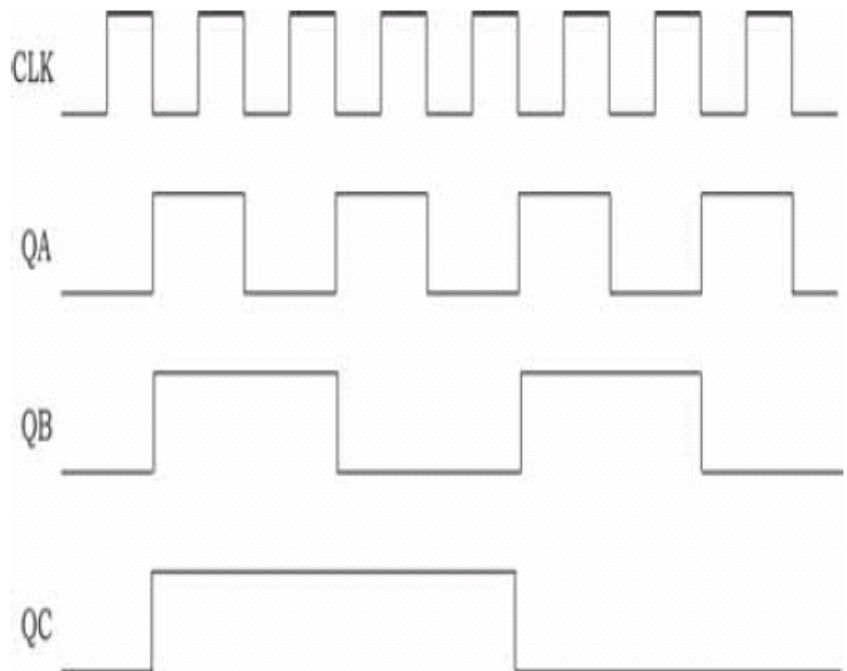
3-bit Asynchronous up counter			
Clock	QC	QB	QA
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0



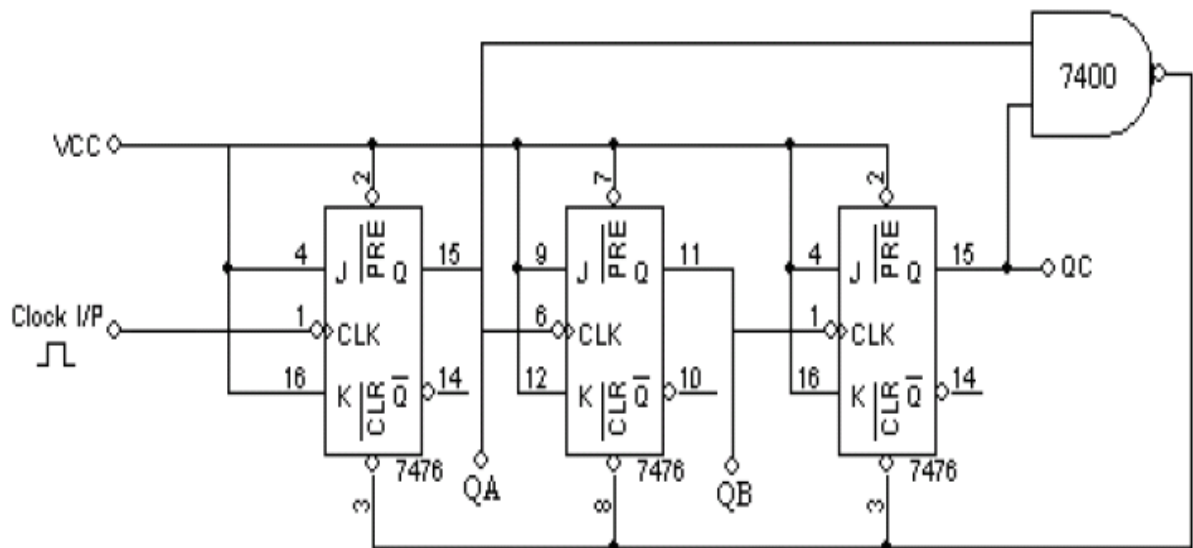
Circuit Diagram: - 3-Bit Asynchronous Down Counter



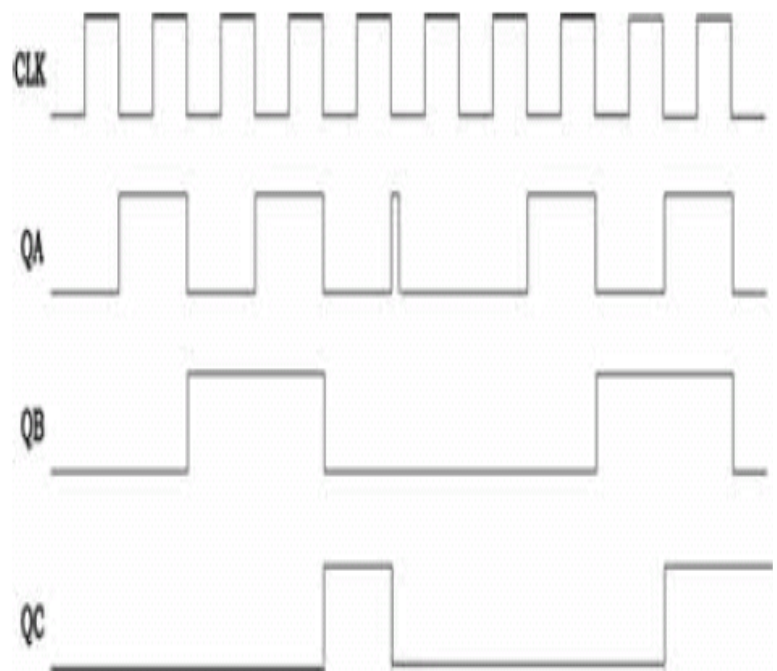
3-bit Asynchronous down counter			
Clock	QC	QB	QA
0	1	1	1
1	1	1	0
2	1	0	1
3	1	0	0
4	0	1	1
5	0	1	0
6	0	0	1
7	0	0	0
8	1	1	1
9	1	1	0



Mod 5 Asynchronous Counter:-



Mod 5 Asynchronous counter			
Clock	QC	QB	QA
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	0	0	0



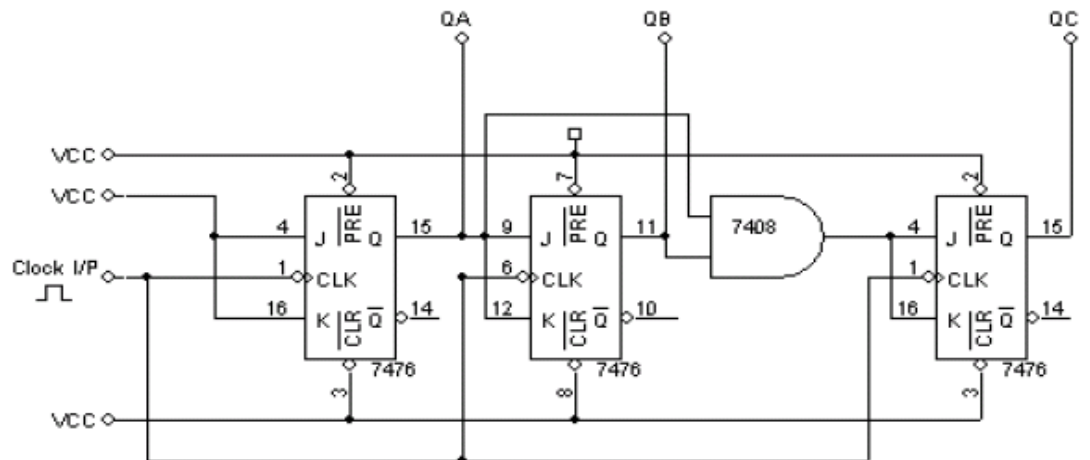
Rating: _____

Aim: - Realization of 3-bit synchronous counter design.

IC 7408, IC 7476, IC 7400, IC 7432 etc.

1. Connections are made as per circuit diagram.
2. Clock pulses are applied one by one at the clock I/P and the O/P is observed at QA, QB & QC for IC 7476.
3. Verify the Truth table .

3-bit Synchronous Counter:-



Truth table :-

Clock	QC	QB	QA
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

1. Logic Probes

A **Logic Probe** is a hand-held pen-like probe used for analyzing and troubleshooting the logical states (Boolean 0 or 1) of a digital circuit. While most are powered by the circuit under test, some devices use batteries. They can be used on either TTL (transistor-transistor logic) or CMOS (complementary metallic oxide semiconductor) integrated circuit devices.



There are usually three differently-colored LEDs on the probe's body:

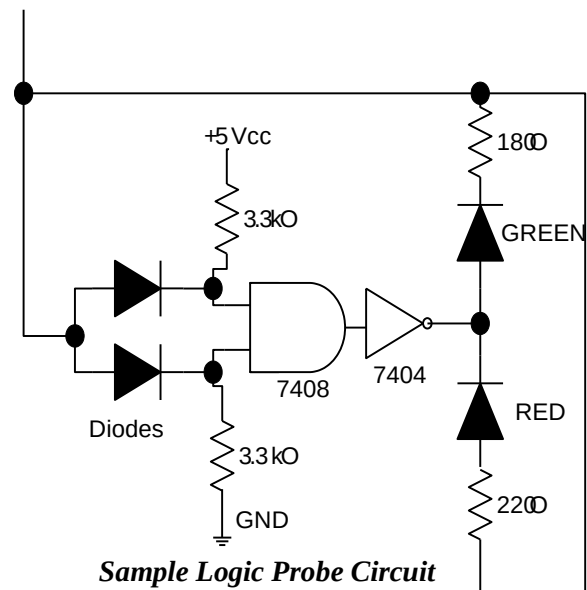
- Red and green LEDs indicate high and low states respectively
- An amber LED indicates a pulse

When the logic probe is either connected to an invalid logic level (a fault condition or a tri-stated output) or not connected at all, none of the LEDs lights up. A logic probe is a cheap, versatile and convenient digital test instrument, but can test only a single signal at a time.

Test **Problem** Probes

Design a simple logic probe circuit that will be able to present the two (2) logic states of a logic gate such as the AND, OR, & NOT.

It must indicate the logic state of the node of any TTL logic circuit. To do that, we have to supply the probe with the same power of the circuit that we want to analyze: same V_{CC} and same GND. To check the logic level, we must connect the "Test" wire of the probe to the desired node of the circuit that we want to check.



Required Reports

1. IC Diagram of the circuit.
2. Description of the project.
3. Description of the Materials used.
4. Discussion on the procedure of implementation
5. Actual implementation of the project using RECYCLED materials.

Date of Submission: _____

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2. Game Show Buzzers

The attraction of any game show depends on the visual effects. The way in which they present different rounds makes the game shows effective. The good Quiz show always contains one or more buzzer rounds.

The buzzer makes a round of a Quiz show 'fastest finger first '. The active buzzer makes the team to first respond to the question. A common question will be asked for all the teams in these rounds. The team which presses the buzzer first gets first chance to answer. To avoid confusion of the teams to be answered when one or more team presses the buzzer almost simultaneously, buzzer circuits are used.

Problem Description:

Create a Logic Circuit that will function like a Game Show Buzzer. This circuit will allow up to four players to compete. Led lights show for whoever pushed the button first. Extend the push buttons from the circuit board with wires and your own momentary switches. Circuit must be reset via the quiz master push button to start over again.

In game shows it is also known as a "lockout system" because when one person signals ("buzzes in"), all others are locked out from signaling. Several game shows have large buzzer buttons which are identified as "plungers". The buzzer is also used to signal wrong answers and when time expires on many game shows.

Required Reports

1. Description of the project.
2. Schematic and IC Diagram of the circuit.
3. Description of the Materials used.
4. Discussion on the procedure of implementation
5. Actual implementation of the project using breadboards.

Date of Submission: _____

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3. Excess three (3) Code Converters

Excess-3 binary-coded decimal (**XS-3**), also called **biased** representation or **Excess- N** , is a numeral system used on some older computers that uses a pre-specified number N as a biasing value. It is a way to represent values with a balanced number of positive and negative numbers. In XS-3, numbers are represented as decimal digits, and each digit is represented by four bits as the BCD value plus 3 (the "excess" amount):

- The smallest binary number represents the smallest value. (i.e. 0 – Excess Value)
- The greatest binary number represents the largest value. (i.e. $2^{N+1} - \text{Excess Value} - 1$)

Decimal	Binary	Decimal	Binary	Decimal	Binary	Decimal	Binary
-3	0000	1	0100	5	1000	9	1100
-2	0001	2	0101	6	1001	10	1101
-1	0010	3	0110	7	1010	11	1110
0	0011	4	0111	8	1011	12	1111

To encode a number such as 127, then, one simply encodes each of the decimal digits as above, giving (0100, 0101, 1010). The primary advantage of XS-3 coding over BCD coding is that a decimal number can be nines' complemented (for subtraction) as easily as a binary number can be ones' complemented by just invert all bits.

Adding Excess-3 works on a different algorithm than BCD coding or regular binary numbers. When you add two XS-3 numbers together, the result is not an XS-3 number. For instance, when you add 1 and 0 in XS-3 the answer seems to be 4 instead of 1. In order to correct this problem, when you are finished adding each digit, you have to subtract 3 (binary 11) if the digit is less than decimal 10 and add three if the number is greater than or equal to decimal 10 (thus causing the number to wrap) **Problem**

Description:

Create a Logic Circuit that will accept 4-bit numbers and will produce an output equal to 4-bit Excess 3. Use LEDs to represent the Output in Binary Numbers.

Required Reports

1. Schematic and IC Diagram of the circuit.
2. Description of the project.
3. Description of the Materials used.
4. Discussion on the procedure of implementation
5. Actual implementation of the project using breadboards.

Date of Submission: _____

4. Farm Water Controller

a circuit that will activate a water tank controller if three or more from the five tanks in a farm lot is filled with water. This controller circuit will follow the truth table shown below:

Tank A	Tank B	Tank C	Tank D	Tank E	(Controller) Output
0	0	0	0	0	0
0	0	0	0	1	0
0	0	0	1	0	0
0	0	0	1	1	0
0	0	1	0	0	0
0	0	1	1	0	0
0	0	1	0	1	0
0	0	1	1	0	0
1	0	1	1	1	1
::	::	::	::	::	::
1	1	0	0	0	0
1	1	0	0	1	1
1	1	0	1	0	1
1	1	0	1	1	1
1	1	1	0	0	1
1	1	1	0	1	1
1	1	1	1	0	1
1	1	1	1	1	1

Legend:

Input:

1 = Water Tank Full 0 = Water Tank Empty

Output:

1 = Water Released 0 = Water Released

Required Reports

1. IC Diagram of the circuit.
2. Description of the project.
3. Description of the Materials used.
4. Discussion on the procedure of implementation
5. Actual implementation (using LEDs) of the project using RECYCLED materials (less Water).

Date of Submission: _____

5.Binary to HEX Letter Converter

a circuit that will convert the input binary number into a Hexadecimal letter using the 7-segment display. This circuit will follow the truth table shown below:

A	B	C	D	(7-segment) Output
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	A
1	0	1	1	B
1	1	0	0	C
1	1	0	1	D
1	1	1	0	E
1	1	1	1	F

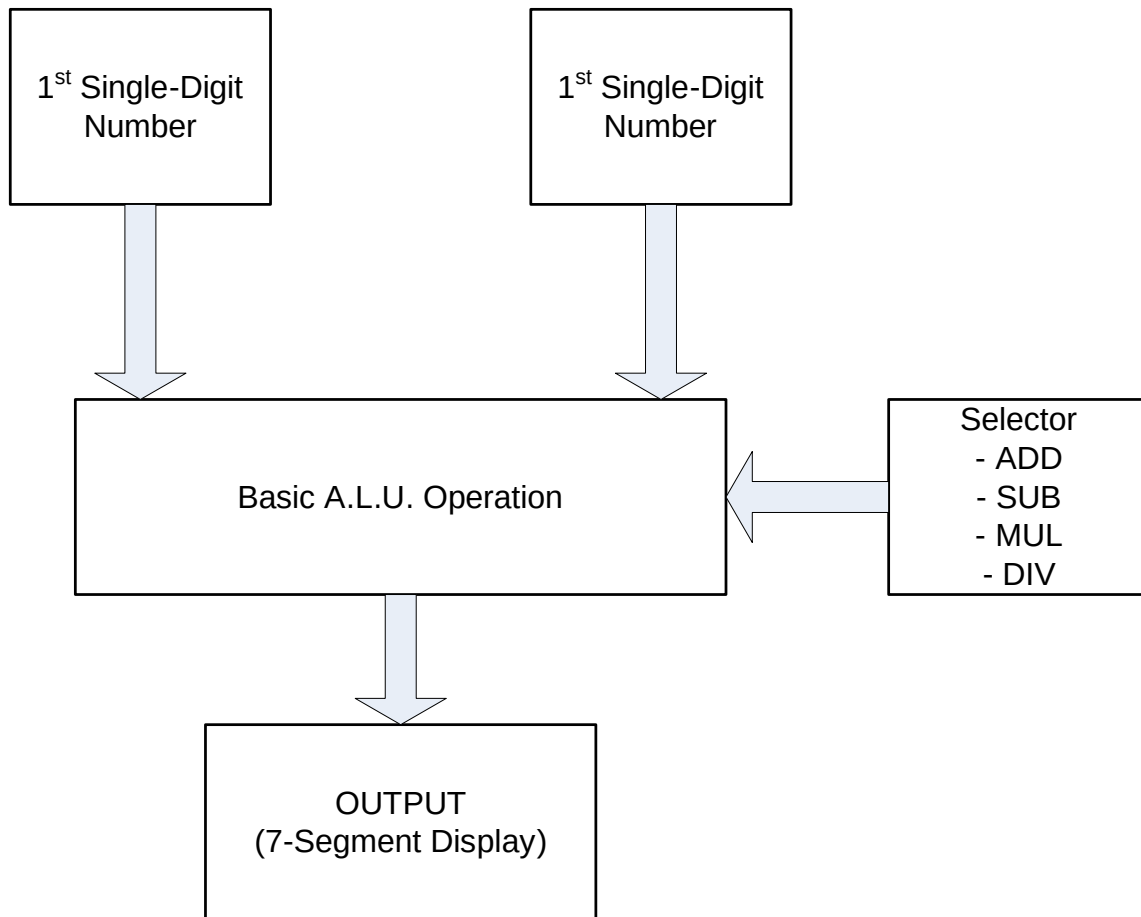
Required Reports

1. IC Diagram of the circuit.
2. Description of the project.
3. Description of the Materials used.
4. Discussion on the procedure of implementation
5. Actual implementation (using LEDs) of the project using RECYCLED materials (less Water).

Date of Submission: _____

6. Basic Arithmetic Logic Unit

a circuit that will add, subtract, multiply & Divide two single- digit numbers and will produce an output SUM, DIFFERENCE, PRODUCT, & QUOTIENT of the two single- digit numbers. This circuit will follow the block diagram shown below:



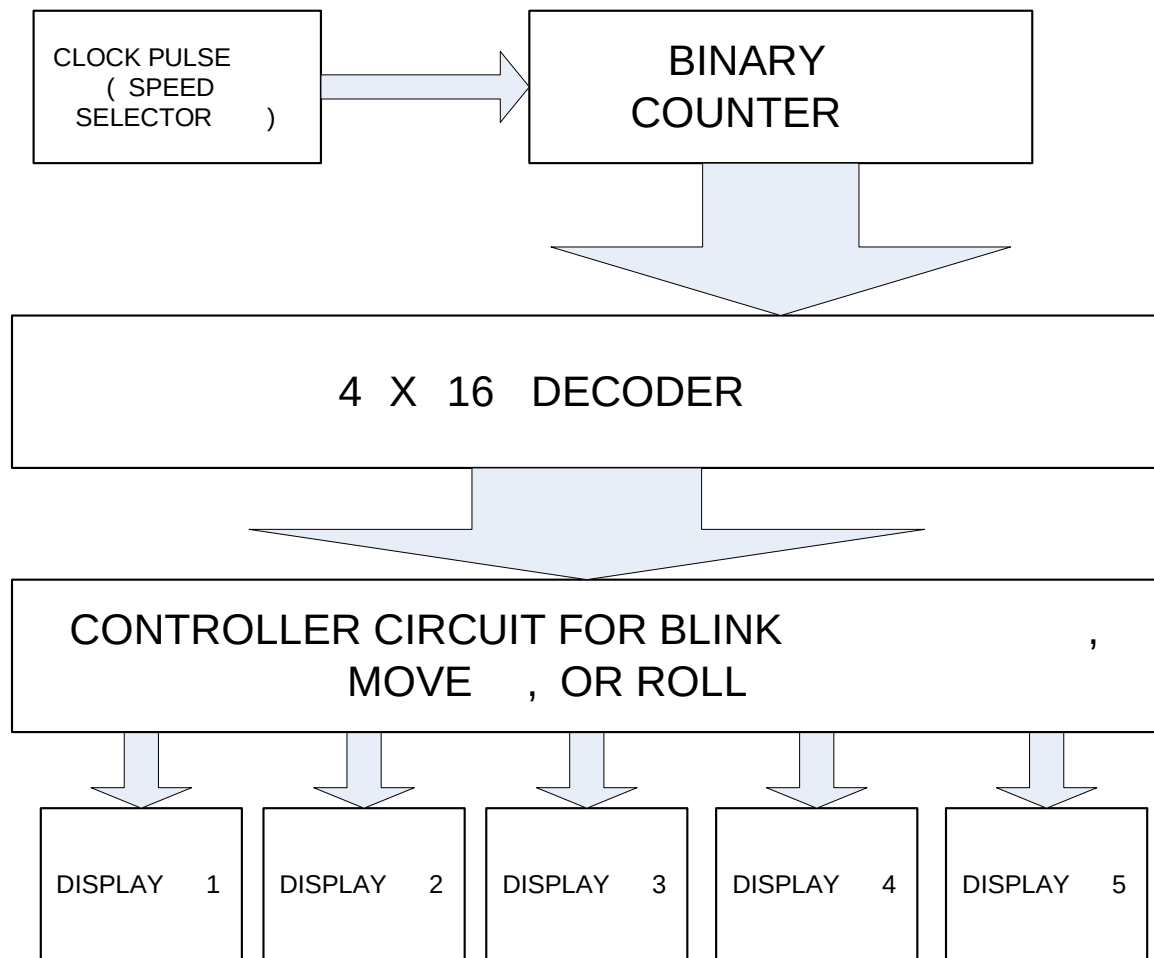
Required Reports

1. IC Diagram of the circuit.
2. Description of the project.
3. Description of the Materials used.
4. Discussion on the procedure of implementation
5. Actual implementation (using LEDs) of the project using RECYCLED materials.

Date of Submission: _____

7.Moving Text Message

a circuit that will display a text and will make it either Blinking, Move – to – Left, Move – to – Right, Roll – to – the - Right, or Roll – to – the - Left This circuit will follow the block diagram shown below:



Required Reports

1. IC Diagram of the circuit.
2. Description of the project.
3. Description of the Materials used.
4. Discussion on the procedure of implementation
5. Actual implementation (using LEDs) of the project using RECYCLED materials.

Date of Submission: _____

Home Project #6

Binary to HEX Letter Converter

Problem

Design a circuit that will convert the input binary number into a Hexadecimal letter using the 7-segment display. This circuit will follow the truth table shown below:

A	B	C	D	(7-segment) Output
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	A
1	0	1	1	B
1	1	0	0	C
1	1	0	1	D
1	1	1	0	E
1	1	1	1	F

Required Reports

1. IC Diagram of the circuit.
2. Description of the project.
3. Description of the Materials used.
4. Discussion on the procedure of implementation
5. Actual implementation (using LEDs) of the project using RECYCLED materials (less Water).

Date of Submission: 1 week after this experiment.